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CINECA AGENTIC PLATFORM

Enterprise-Grade AI Agent Orchestration Platform Connecting Multi-Tenant Architecture, MCP Tools, and Secure Graph Querying

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Abstract

Large Language Models (LLMs) can reason over text, but deploying them as autonomous agents inside an enterprise raises problems that no single existing framework solves together: multi-tenant data isolation, auditable tool execution, reliable multi-step planning, and safe access to knowledge graphs. At CINECA, Italy's national supercomputing centre, researchers who need to query graph-structured bioinformatics data cannot write Cypher and must rely on developers, creating delays and bottlenecks.

This thesis presents the *Cineca Agentic Platform* (CAP), a full-stack system that closes this gap. CAP pairs a three-layer architecture with a four-phase orchestration engine: intent classification, LLM-based plan generation, step-by-step execution (tool calls or graph queries), and safe response finalization with personally-identifiable-information (PII) scrubbing. Two execution workflows serve different needs: synchronous *Agent Runs* for interactive chat and durable asynchronous *Jobs* with real-time Server-Sent Events (SSE) streaming. A native Model Context Protocol (MCP) runtime governs 34 tools across seventeen categories with schema validation, role-based access control (RBAC), and audit logging. A Natural-Language-to-Cypher pipeline subjects every generated query to six safety layers before executing it on a graph database. Security is built in via an eight-stage middleware stack covering OpenID Connect / JSON Web Token authentication, RBAC, rate limiting, input/output guards, and tenant-scoped filtering. The architecture is LLM-agnostic with circuit breakers and provider fallback.

The platform exposes 76 API endpoints and 34 MCP tools, validated by more than 3,000+ automated tests spanning unit, integration, security, and end-to-end scenarios. A comparison against ten contemporary platforms shows that CAP is the only solution that natively combines full-stack deployment, durable agentic orchestration, an MCP tool ecosystem, tenant-aware graph querying, and integrated observability.

Keywords: AI Agents, Large Language Models, Orchestration, Model Context Protocol, Graph Database, Multi-tenancy, Enterprise Security, Natural Language to Cypher

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List of Acronyms and Glossary

Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
CAP	CINECA Agentic Platform
CINECA	Consorzio Interuniversitario per il Calcolo Automatico
JWT	JSON Web Token
LLM	Large Language Model
MCP	Model Context Protocol
NL	Natural Language
OIDC	OpenID Connect
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
REST	Representational State Transfer
SSE	Server-Sent Events

Glossary

Audit	An immutable record of a system action for compliance and traceability.
Job	A durable, asynchronous long-running agent task with progress tracking.
Principal	An authenticated user or service identity within the platform.
Run	A single synchronous execution of an agent conversation or task.
Step	An individual action or tool call within a run.
Tenant	An isolated organizational unit with its own data, users, and policies.
Tool	A registered capability (function) that an agent can invoke via MCP.

Chapter 1

Introduction

This chapter motivates the need for an enterprise-grade AI agent orchestration platform, states the concrete problem that this thesis addresses, and summarises the solution and its key results. Section 1.1 describes the production gap between research-oriented agent frameworks and what enterprise environments actually require. Section 1.2 isolates the five deficiencies that make naïve chatbot deployments unsuitable for a multi-tenant supercomputing centre. Sections 1.3–1.4 define the objectives, scope, and high-level design of the Cineca Agentic Platform. Sections 1.5–1.6 list the measurable contributions and preview the evaluation outcomes. Section 1.7 provides a chapter-by-chapter guide to the remainder of this document.

1.1 Context and motivation

Large Language Models (LLMs) such as GPT-4 and LLaMA have demonstrated strong capabilities in natural-language understanding, reasoning, and generation. A natural next step is to embed these models inside *AI agents*: software entities that perceive their environment, decompose a goal into sub-tasks, invoke external tools, and synthesise a final answer across multiple reasoning steps. The promise is compelling—researchers could query complex datasets in plain language, operators could trigger maintenance workflows conversationally, and analysts could cross-reference graph-structured knowledge without writing a single line of Cypher, SQL, or shell script.

In practice, however, a wide gap separates laboratory demonstrations of such agents from systems that can be deployed safely inside an enterprise. This gap, referred to throughout this thesis as the *production gap*, has three dimensions. First, research-oriented frameworks such as LangChain and LangGraph are libraries rather than complete stacks: they provide agent abstractions and prompt-chaining utilities but leave infrastructure concerns—authentication, multi-tenant data isolation, durable job execution, observability—to the adopter. Second, basic chatbot deployments can handle simple single-turn question answering but cannot plan multi-step workflows, invoke governed tools, or survive a server restart mid-execution. Third, in a survey of ten representative platforms in the AI orchestration space (introduced in Chapter 2 and discussed quantitatively in Chapter 5), we did not find a platform that natively combines agentic orchestration with safe access to a graph database—the

exact capability that CINECA’s bioinformatics researchers need.

CINECA (Consorzio Interuniversitario per il Calcolo Automatico) [5] is Italy’s national supercomputing centre, serving a large consortium of universities and research institutions. Among its many activities, CINECA maintains a knowledge graph that captures bioinformatics entities—users, institutions, computational tasks, data files, genes, proteins, pathways, and their relationships. Cypher is the query language used to interrogate such graphs. Researchers who wish to explore this graph must currently write Cypher queries or ask a developer to write them on their behalf. Both paths introduce delays: the first requires query-language expertise that most domain scientists lack; the second creates a developer bottleneck that slows down exploratory analysis.

The confluence of these observations—the proliferation of capable LLMs, the inadequacy of library-only agent frameworks for enterprise use, and CINECA’s concrete need for safe natural-language access to graph data—motivates the work presented in this thesis. The central question is whether a single, integrated platform can close the production gap by combining full-stack deployment, durable agentic orchestration, a governed tool ecosystem, tenant-aware graph querying, and built-in observability.

1.2 Problem statement

Deploying an LLM-powered agent inside a multi-tenant enterprise environment exposes five concrete deficiencies that existing frameworks and naïve chatbot prototypes do not address together.

No multi-step planning or durable execution. A basic chatbot processes one prompt at a time. If a task requires several intermediate actions—for example, classifying the user’s intent, generating a graph query, executing it, and summarising the results—the chatbot has no built-in mechanism to plan, track, or recover those steps. If the hosting process crashes, any in-flight work is lost.

No governed tool ecosystem. Agentic operation becomes useful only when the agent can invoke external tools—cache lookups, graph queries, data exports, model management operations. Without a registry that validates each tool’s input schema, enforces role-based access control (RBAC), and logs every invocation, tool use becomes an unaudited attack surface. RBAC is the practice of assigning permissions not to individual users but to named roles (e.g., admin, operator, user, viewer) so that access policies remain manageable as the user base grows.

No multi-tenant data isolation. A *tenant* is an organisational unit—a research group, a department, or an entire institution—whose data, configuration, and policies must be isolated from those of every other tenant sharing the same platform. Research-oriented agent frameworks commonly assume a single-tenant deployment. In a centre such as CINECA, where many institutions share the same infrastructure, the absence of tenant-scoped filtering means that one group could inadvertently—or intentionally—access another group’s data.

No safe graph querying. Translating a natural-language question into a Cypher query and executing it on a live graph database is inherently risky. Without validation, an LLM might generate a write query that mutates production data, a deeply nested traversal that exhausts memory, or a query that crosses tenant boundaries. This pipeline, called *NL-to-Cypher* throughout this thesis, requires multiple safety layers: syntax checking, read-only enforcement, tenant filtering, depth and timeout limits, and result-size caps.

No integrated security, audit, and observability. Enterprise deployments require authentication (who is calling?), authorisation (what are they allowed to do?), audit logging (what did they do?), rate limiting (how often can they do it?), and observability (how is the system behaving?). These concerns cut across every layer of the stack. Bolting them onto a library-only framework after the fact is fragile and error-prone.

These five deficiencies are not independent; they compound. An agent that can invoke tools but cannot verify the caller’s identity is a security liability. A pipeline that can translate natural language to Cypher but cannot enforce tenant boundaries is a data-isolation failure. The thesis therefore poses a single, integrated research problem: *Can a purpose-built platform close the production gap by addressing all five deficiencies within a unified architecture?*

1.3 Objectives and scope

The primary objective of this thesis is to design, implement, and evaluate a full-stack AI agent orchestration platform—the *Cineca Agentic Platform* (CAP)—that is intended to close the production gap identified in Section 1.2. The work targets four measurable goals:

1. Provide two execution workflows—synchronous *Agent Runs* for interactive chat and durable asynchronous *Jobs* with real-time progress streaming via Server-Sent Events (SSE)—so that both quick questions and long-running tasks are supported. An *Agent Run* (or *run*) is a single execution of the orchestrator in response to a user prompt; a *Job* is a durable, queue-backed unit of work whose progress survives process restarts.
2. Implement a governed tool ecosystem based on the Model Context Protocol (MCP), a specification that standardises how AI agents discover and invoke external capabilities. Every tool is registered with a JSON schema, protected by RBAC, and its invocations are audit-logged.
3. Deliver a six-stage NL-to-Cypher pipeline that translates natural-language questions into validated, tenant-scoped Cypher queries and executes them safely against a graph database.
4. Integrate security, multi-tenancy, and observability as first-class architectural concerns rather than afterthoughts.

The scope of the thesis encompasses the platform’s architecture, its four-phase orchestration engine, the MCP tool runtime, the NL-to-Cypher subsystem, the security middleware stack, the data layer, and the observability infrastructure. Two user interfaces—a chat application for end users and an administrative control panel for operators—are included as integral parts of the full-stack design.

Certain topics fall outside this scope. CAP is not a general-purpose LLM training or fine-tuning framework; it consumes pre-trained models via provider adapters. It is not a replacement for HPC job schedulers such as SLURM; its job system handles agent-oriented tasks, not arbitrary compute-resource scheduling. Formal verification of generated Cypher queries and adversarial robustness testing of the LLM components are identified as areas for future work rather than claims of this thesis.

1.4 Proposed approach

CAP adopts a three-layer architecture—presentation, backend services, and data infrastructure—with cross-cutting modules for security, configuration, and observability. At the presentation layer, a chat interface serves end users while an administrative control panel serves operators. Both communicate with the backend through a versioned REST API (Representational State Transfer Application Programming Interface) behind a reverse proxy.

The backend centres on a four-phase *orchestration engine*. When a user submits a prompt, the engine first classifies the intent into one of six categories: conversational chat, graph query, security inquiry, administrative action, a potentially dangerous operation, or an explain-only analysis. It then generates a plan—a sequence of to-do items—using an LLM. Each to-do item is executed as a *step*, which may involve calling an LLM provider, invoking an MCP tool, or running the NL-to-Cypher pipeline. A *step* is the smallest auditable unit of work inside a run: it records its action, input, output, and latency. Finally, the engine assembles the results, applies personally identifiable information (PII) scrubbing and output-safety checks, and returns a normalised response.

Two execution workflows share the same core orchestration logic. The synchronous path (Agent Runs) executes within the API process and is polled by the client. The asynchronous path (Jobs) enqueues work onto a queue in the in-memory store, where a dedicated worker process picks it up, executes the same orchestrator, and streams progress events to the client via SSE. The asynchronous path is designed to survive process restarts and to scale horizontally by adding workers.

The data layer uses three complementary stores: a relational database as the durable control plane (tenants, runs, steps, jobs, tool definitions, audit logs), an in-memory store for the low-latency data plane (caching, job queues, rate-limit counters, session state), and a graph database that hosts the bioinformatics domain model and serves as the target of the NL-to-Cypher pipeline.

Security is enforced through a multi-stage middleware stack applied to every API route before the request reaches business logic. The stages cover cross-origin policy, distributed tracing, token-based authentication with RBAC enforcement, rate limiting, tenant resolution, input validation and injection prevention, output

sanitisation, and centralised error handling. Because the same middleware chain is applied to all API routes, security and tenant checks are enforced consistently across the platform. Chapter 5 reports dedicated negative and positive security tests exercising token validation, RBAC, tenant isolation, rate limiting, injection prevention, and PII scrubbing.

The platform is designed to be LLM-agnostic. Provider adapters abstract the differences between cloud-hosted, enterprise, and locally self-hosted models, and a resilience layer adds automatic retries, provider fallback chains, and failure isolation so that a transient error in one provider does not cause the in-flight request to fail.

1.5 Contributions

This thesis makes the following contributions, each of which is designed to be independently testable:

- A **full-stack, enterprise-grade AI agent platform** comprising two user interfaces, a versioned API layer, and an asynchronous worker subsystem, deployable as a single containerised stack.
- A **four-phase orchestration engine** that classifies intent, generates a multi-step plan, executes each step through LLM calls, tool invocations, or graph queries, and finalises the response with safety checks.
- A **governed MCP tool runtime** hosting 34 tools across seventeen categories, each registered with a JSON input schema, protected by RBAC and per-tool rate limits, and audit-logged on every invocation.
- A **six-stage NL-to-Cypher pipeline** that translates natural-language questions into tenant-scoped, safety-validated Cypher queries executed against a graph database.
- A **multi-stage security middleware stack** providing token-based authentication, four-role RBAC, rate limiting, tenant-scoped data isolation, input and output guards, and PII scrubbing.
- A **comparative evaluation** against ten contemporary platforms—Temporal, Argo Workflows, LangGraph, OpenAI Agents SDK, Semantic Kernel, LlamaIndex, Haystack, n8n, Windmill, and Langfuse—indicating that, among the surveyed platforms, CAP is the only one that natively combines full-stack deployment, durable agentic orchestration, a governed tool ecosystem, tenant-aware graph querying, and integrated observability.

1.6 Results preview

The platform was validated through more than 3,000 automated tests spanning unit, integration, security, and end-to-end scenarios, together with a structured capability comparison against ten contemporary platforms in the AI orchestration space. The principal outcomes are summarised below.

Functional coverage. Both execution workflows—synchronous Agent Runs and durable asynchronous Jobs—were exercised end-to-end. The four-phase orchestration engine handles six intent categories (chat, graph, security, admin, dangerous, explain), and the NL-to-Cypher pipeline was tested with representative bioinformatics queries over a graph schema comprising fourteen node types and four relationship types.

Security validation. The test suite includes dedicated security tests covering token validation, RBAC enforcement across four roles (admin, operator, user, viewer), tenant isolation, rate limiting, injection prevention, and PII scrubbing. The middleware stack is exercised by dedicated tests that cover enforcement of the security stages across the API surface, including negative cases (Chapter 5).

Comparative positioning. A capability-by-capability comparison against ten platforms—including Temporal (workflow durability), LangGraph (agent loops with state checkpointing), and Langfuse (LLM tracing)—indicates that, among the surveyed platforms, CAP is the only one that achieves full marks across eight evaluation dimensions: full-stack deployment, orchestration durability, agent planning loop, tool ecosystem, security and governance, graph NL-to-Cypher support, observability, and LLM-agnostic design.

LLM-agnostic operation. The adapter and resilience layer was exercised with three provider families—cloud-hosted, enterprise, and locally self-hosted—demonstrating that the platform can switch providers at the request, session, or tenant level without code changes.

1.7 Thesis roadmap

The remainder of this document is organised as follows.

Chapter 2 reviews the background material necessary to understand the design decisions: agentic systems and tool use, orchestration and workflow engines versus agent frameworks, tool-governance challenges, and natural-language interfaces to graph databases. It closes with a gap analysis that positions CAP relative to the state of the art.

Chapter 3 presents the platform itself. It opens with requirements and design goals, describes the three-layer architecture and core concepts, details the dual-workflow execution model with an end-to-end walkthrough, and then covers the orchestration engine, the MCP tool ecosystem, the NL-to-Cypher subsystem, the data layer, the security model, observability, and design trade-offs.

Chapter 4 defines the evaluation methodology: experimental setup, evaluation questions mapped to requirements, metrics, test protocols, and baseline selection.

Chapter 5 reports and discusses the results—functional, performance, reliability, security, and comparative—and identifies limitations and threats to validity.

Chapter 6 recaps the problem, the solution, and the key findings, while Chapter 7 outlines concrete next steps including workflow unification, strengthened autonomy, and deployment hardening.

In summary, this introduction has framed the production gap that separates research-oriented agent frameworks from enterprise-ready deployments, identified five concrete deficiencies that compound in multi-tenant environments, and outlined how the Cineca Agentic Platform is designed to address them. The next chapter provides the technical background needed to evaluate these design choices.

Chapter 2

Related Works and Background

This chapter reviews the technical foundations and prior work that inform the design of the Cineca Agentic Platform. It covers four domains: the emergence of Large Language Model (LLM) based agents and their reliance on external tools (Section 2.1), the spectrum of orchestration engines and agent frameworks with their respective strengths and limitations (Section 2.2), the growing need for governed tool ecosystems and the Model Context Protocol that addresses it (Section 2.3), and the challenge of translating natural-language questions into safe, tenant-scoped graph-database queries (Section 2.4). The chapter closes with a gap analysis (Section 2.5) that synthesises what existing systems do not provide together and positions the contribution of this thesis.

2.1 Agentic systems and tool-use

2.1.1 From language models to agents

A Large Language Model is a neural network, typically based on the Transformer architecture introduced by Vaswani et al. [31], that is trained on a large text corpus to predict the next token in a sequence. Scaling studies have shown that model performance improves predictably with increases in parameter count, dataset size, and compute budget [15]. Beyond a certain scale, LLMs exhibit *emergent capabilities* that were not explicitly programmed: in-context learning (performing new tasks from examples provided in the prompt), chain-of-thought reasoning (decomposing a problem into intermediate steps [32]), and—most relevant for this thesis—*tool use*, the ability to generate structured function calls that invoke external services [29].

These capabilities have given rise to *AI agents*: software entities that perceive their environment through observations, reason about goals and constraints with the help of an LLM, and take actions—such as invoking a database query or calling an external API—to achieve a specified objective. Unlike a basic chatbot, which processes one prompt and returns one response, an agent operates in a loop: it observes the result of its last action, updates its internal state, and decides on the next action until the goal is met or a termination condition is reached.

2.1.2 Agent reasoning patterns

Several architectural patterns have been proposed for structuring this loop. The *ReAct* framework [35] interleaves reasoning traces with action execution: at each step the agent (i) thinks (generates a natural-language explanation of its plan), (ii) acts (selects and invokes a tool), and (iii) observes (incorporates the tool’s output into context). An alternative is the *plan-and-execute* paradigm, in which the agent first generates a complete plan—a sequence of sub-tasks—and then executes each sub-task in order, re-planning only when an intermediate step fails. More advanced approaches such as Tree-of-Thought [34] maintain multiple reasoning branches and use search algorithms to find successful paths, at the cost of significantly higher computational expense.

In practice the choice of reasoning pattern depends on the deployment context. Interactive scenarios, where a user expects low-latency responses, favour single-pass ReAct loops. Batch or analytical scenarios, where correctness outweighs speed, can afford the overhead of plan-and-execute or tree-search strategies. The platform described in this thesis adopts a ReAct-style loop for its synchronous path and a plan-based approach for its asynchronous path, a trade-off discussed in Chapter 3.

2.1.3 The role of tool use

Tool use transforms a language model from a text generator into an agent that can affect the external world. When an LLM is provided with a set of tool definitions—each described by a name, a natural-language description, and a JSON schema specifying the expected input parameters—it can recognise when a tool invocation is appropriate, generate a structured call with the required arguments, and incorporate the tool’s output into its subsequent reasoning. OpenAI’s function-calling convention [24], introduced in June 2023, established a de facto standard for such definitions and has since been adopted by many other providers.

Tool categories in agent systems span information retrieval (web search, database queries), computation (code interpreters, calculators), action execution (API calls, file operations), and memory management (reading from and writing to persistent storage). As the number and power of available tools grow, so does the risk of misuse. An agent that can query a database could also mutate it; an agent that can call an external API could exfiltrate sensitive data. In a multi-user, multi-tenant environment such as a supercomputing centre, these risks are not hypothetical. Three governance requirements therefore accompany any serious tool ecosystem: *authorisation* (which user or role may invoke which tool), *validation* (are the tool’s inputs well-formed and safe?), and *auditability* (is every invocation logged for compliance and forensics?). These concerns motivate the tool governance model discussed in Section 2.3 and the platform’s MCP tool runtime described in Chapter 3.

2.1.4 Challenges in production deployment

Deploying LLM-based agents in production raises challenges beyond those of conventional web services. First, LLMs can produce incorrect or inconsistent outputs, particularly for edge cases, so the infrastructure must detect failures, retry with backoff, and fall back to alternative providers when a primary model is unavailable.

Second, LLM inference is computationally expensive and introduces variable latency; enterprise deployments must monitor response times, track token consumption for cost control, and choose appropriately sized models for different tasks. Third, agents with tool access can potentially execute sensitive operations, which demands authentication, authorisation, and audit trails at every layer. Fourth, understanding agent behaviour in production requires comprehensive observability: request-level metrics, structured logs with correlation identifiers, and distributed traces that follow a request from ingress through the orchestrator, the LLM provider, and any tool invocations.

These challenges—reliability, cost, security, and observability—cannot be addressed by a library alone. They require an integrated platform that treats them as first-class architectural concerns, a point that motivates the survey of existing frameworks in the next section.

2.2 Orchestration and workflow engines versus agent frameworks

The need to coordinate multi-step, tool-augmented agent work has produced a diverse ecosystem of frameworks, libraries, and platforms. They differ along three axes that are important for enterprise adoption: the *abstraction level* (composable primitives, agent runtimes, or full-stack platforms), the *deployment model* (in-process library, standalone service, or managed cloud), and the *scope of concerns* they address (prompting, tool invocation, persistence, security, observability). This section surveys the most prominent alternatives, grouped into workflow orchestration engines, agent-oriented frameworks, and full-stack automation platforms.

2.2.1 Workflow orchestration engines

Temporal [30] is a leading open-source workflow orchestration engine that provides robust *durable execution*: workflows survive process crashes, network outages, and even infrastructure migrations because every state transition is persisted and can be replayed. Temporal offers polyglot SDKs (Go, Java, Python, TypeScript, among others) and automatic retry logic with configurable back-off. However, Temporal is an engine, not an agent platform: although it ships a monitoring dashboard for workflow visibility, it does not provide an end-user agent-interaction UI (e.g., chat or administration), and it has no built-in agentic planning loop and no graph-database integration. Building an agent on Temporal would require implementing intent classification, tool governance, tenant isolation, and observability from scratch.

Argo Workflows [3] takes a container-native approach: each workflow step runs as a Kubernetes pod, leveraging Kubernetes for scheduling, resource management, and security. Argo supports directed acyclic graphs for modelling complex step dependencies and provides a web interface for monitoring. Its limitations mirror Temporal’s from an agentic perspective: there is no agent loop, no natural-language interface, and access control relies entirely on Kubernetes RBAC, which operates at the namespace level rather than at the level of individual tools or tenants.

Both engines excel at what they were designed for—durable, infrastructure-level

workflow coordination—but they do not address the higher-level concerns of LLM orchestration, tool governance, or multi-tenant data isolation.

2.2.2 Agent-oriented frameworks

LangChain [4] is arguably the most widely adopted framework for building LLM-powered applications. It provides composable abstractions—models, prompts, chains, agents, memory, retrievers, and tools—with integrations for many LLM providers, vector stores, and external services. Its primary strength is rapid prototyping: a developer can assemble a ReAct agent with web-search tools in a few lines of code. For enterprise deployment, however, LangChain presents significant gaps. It provides no built-in authentication, authorisation, or audit logging; multi-tenant isolation must be implemented by the application developer; and because it is an in-process library, running it as a standalone service with its own API, persistence, and health probes requires substantial additional infrastructure.

LangGraph [16], a newer addition to the LangChain ecosystem, addresses some of these limitations by introducing explicit state machines for agent control flow, durable state checkpointing, human-in-the-loop approval patterns, and streaming of intermediate results. LangGraph represents a step toward production readiness, but it still relies on external systems for authentication, multi-tenancy, and operational concerns.

LlamaIndex [19] focuses on the data side of LLM applications, offering sophisticated connectors, indexes, and query engines for retrieval-augmented generation (RAG). Its agent support is less mature than that of agent-focused frameworks, and, like LangChain, it is a library without built-in production infrastructure.

Microsoft’s Semantic Kernel [21] takes a plugin-oriented approach with SDKs for C#, Python, and Java, providing planners that generate multi-step execution plans from high-level goals. It integrates tightly with the Azure ecosystem but offers limited support for multi-tenancy and requires the host application to provide its own security layer.

The OpenAI Agents SDK [25], released in March 2025, provides a Python-first framework with a built-in agent loop, tool invocation, guardrails, and multi-agent handoffs. It is designed primarily for OpenAI models and does not include job scheduling, RBAC, tenant isolation, or graph-database support.

A common thread runs through all these frameworks: they excel at the *agent logic* layer—prompt composition, reasoning loops, tool calling—but leave *infrastructure* concerns to the adopter. Authentication, tenant-scoped data access, durable job execution, rate limiting, and production observability must be bolted on after the fact, which increases development effort, introduces integration risk, and makes it difficult to enforce security and governance consistently.

2.2.3 Full-stack automation platforms

A smaller number of platforms do attempt a full-stack approach. n8n [22] is a visual workflow automation tool that offers a web interface, an API, and background execution; however, its agent support is limited and it lacks fine-grained RBAC, graph-database integration, and LLM-native orchestration. Windmill [33] targets

developer-centric automation with scripts and flows, providing built-in authentication and scheduling but no agentic planning loop and no graph-querying capability. Langfuse [17] focuses exclusively on LLM observability—tracing, evaluation, and prompt management—without any workflow execution or agent orchestration.

None of these platforms addresses the complete set of concerns that an enterprise agentic deployment requires: full-stack UI and API, durable orchestration with an agent planning loop, a governed tool ecosystem, multi-tenant security, graph-native knowledge access, and integrated observability. Section 2.5 quantifies this observation through a structured capability comparison.

2.3 Tool ecosystems and governance

As the number of tools available to LLM-based agents grows, so does the need for a *governed* tool ecosystem—one that standardises how tools are discovered, invoked, and audited. This section reviews the governance problem, introduces the Model Context Protocol (MCP) as a standardisation effort, and discusses the enterprise extensions that the protocol alone does not cover.

2.3.1 The governance problem

Each of the agent frameworks surveyed in Section 2.2 defines its own abstraction for tools: LangChain uses a `Tool` class with a `run()` method, Semantic Kernel uses *plugins* with *skills*, and the OpenAI API uses function definitions attached to the model call. This fragmentation means that a tool implemented for one framework cannot easily be reused in another, that tool authors must maintain parallel implementations, and that similar tools may behave differently across frameworks because conventions differ.

Beyond interoperability, enterprise deployments face governance concerns that research-oriented frameworks do not address. In a multi-tenant environment, different tenants may be entitled to different subsets of tools; an “admin” role may invoke tools that a “viewer” role cannot; every invocation must be logged with the caller’s identity, the tool’s inputs, the output, and the execution time; and per-user or per-tenant rate limits must prevent abuse or accidental resource exhaustion. Without a unified registry and invocation layer, enforcing these policies consistently is difficult and error-prone.

2.3.2 The Model Context Protocol

The Model Context Protocol (MCP), introduced by Anthropic in late 2024 [1], is a specification that standardises how AI applications—referred to as *MCP clients*—discover and invoke capabilities exposed by *MCP servers*. The protocol follows a client–server architecture in which communication uses JSON-RPC 2.0 messages over configurable transports (standard I/O for local processes, HTTP with Server-Sent Events for remote services).

A typical MCP interaction proceeds in four phases. First, the client connects to the server and exchanges capability information (*initialisation*). Second, the client requests the list of available tools via a standardised “tools/list” method (*discovery*).

Third, for each tool the client receives a JSON Schema that describes the tool’s input parameters, enabling automatic validation and documentation generation (*schema retrieval*). Fourth, when the LLM decides a tool should be called, the client sends a “tools/call” request with the tool name and arguments, and the server returns the result (*invocation*).

MCP brings three benefits. First, because tool definitions follow a common schema, a tool implemented once can be used by any MCP-compatible client, reducing vendor lock-in and duplicated effort. Second, the client–server separation enables process isolation: tool execution runs in a separate process or even a separate network zone, limiting the blast radius of failures. Third, the standardised invocation interface provides a natural interception point for authentication, logging, and rate limiting.

A growing ecosystem of MCP servers already exists [2], including reference servers for file-system access, GitHub operations, and PostgreSQL queries, as well as community-contributed servers for Neo4j graph databases, Slack, and Google Drive.

2.3.3 Enterprise extensions beyond MCP

MCP standardises *interoperability* but does not prescribe *governance*. The specification does not define how to authenticate callers, how to enforce role-based permissions, how to isolate tool access across tenants, or how to persist audit records. These are precisely the concerns that distinguish a production deployment from a prototype.

An enterprise tool runtime must therefore extend the MCP model in several directions: integrating tool-level authorisation with the platform’s identity and RBAC system, scoping tool visibility and data access to the caller’s tenant, recording every invocation (with caller identity, inputs, outputs, latency, and outcome) in an immutable audit log, and enforcing configurable rate limits per user, per tenant, and per tool. Chapter 3 describes how the platform presented in this thesis implements these extensions while preserving MCP-style invocation patterns.

2.4 Natural-language interfaces to graphs

Graph databases store data as nodes connected by typed, directed relationships—a representation well suited to domains where the connections between entities are as important as the entities themselves. This section introduces the property-graph model and the Cypher query language, reviews approaches to translating natural-language questions into Cypher, and explains why such translation requires multiple safety layers in a multi-tenant deployment.

2.4.1 Property graphs and Cypher

In the property-graph model a *node* represents an entity, carries one or more *labels* that categorise it (for example, *Workflow*, *Dataset*, *User*), and stores attributes as key–value *properties*. A *relationship* is a directed edge between two nodes, carrying a type (such as *DEPENDS_ON* or *PRODUCES*) and its own set of properties. Compared with relational databases, property graphs avoid join-heavy query patterns by representing

relationships explicitly, making relationship traversal a first-class operation rather than a derived one. They also offer schema flexibility (nodes may carry varying properties without migration) and a query idiom that naturally expresses path-based and pattern-matching questions [10, 13].

Cypher is the most widely adopted query language for property graphs [26]. Originally developed for Neo4j [23], it is now supported by multiple databases through the openCypher standard, including Memgraph [20]. Its ASCII-art syntax represents graph patterns visually, making queries readable even to non-experts. Cypher provides clauses for pattern matching (**MATCH**), filtering (**WHERE**), projection (**RETURN**), mutation (**CREATE**, **SET**, **DELETE**), and aggregation.

2.4.2 Natural-language-to-Cypher translation

The central challenge addressed by natural-language-to-Cypher (NL-to-Cypher) systems is: given a question expressed in everyday language and a graph schema describing the available node types, relationship types, and properties, generate a syntactically correct and semantically appropriate Cypher query that answers the question. This challenge has several dimensions. The system must understand the graph schema in order to map nouns and verbs in the question to node labels and relationship types. It must resolve ambiguous references (“the preprocessing workflow”) to specific entities. It must infer implicit relationships and choose appropriate aggregation functions for questions such as “how many workflows use GPU nodes?”. And it must handle path queries of variable length for questions about transitive dependencies or data lineage.

Historically, NL-to-Cypher systems used rule-based methods (high precision on covered patterns, but limited coverage), semantic parsing (formal methods requiring linguistic expertise), or neural sequence-to-sequence models (requiring large training sets for each domain). More recently, LLM-based generation has become the dominant approach because of its flexibility and zero-shot capability: the LLM is prompted with the graph schema, optional examples, and the user’s question, and it generates a Cypher query directly. This approach eliminates the need for domain-specific training data but introduces a new problem—the generated query may be syntactically malformed, semantically incorrect, or unsafe.

2.4.3 Safety requirements for LLM-generated queries

Executing an LLM-generated Cypher query against a live graph database without validation is dangerous for at least three reasons.

First, the LLM may produce a query that *mutates* production data. A question such as “remove all outdated workflows” might cause the model to generate a **DELETE** clause, permanently destroying data. *Read-only enforcement*—rejecting any query that contains **CREATE**, **SET**, **DELETE**, **MERGE**, or **REMOVE** clauses—is therefore essential for analytical use cases.

Second, in a multi-tenant environment the generated query must be *scoped to the caller’s tenant*. A Cypher query that matches all nodes of a given label without a tenant filter could return data belonging to other organisations. Automatic injection

of a tenant predicate (for example, `WHERE n.tenant_id = $tenant`) into every generated query is necessary to prevent cross-tenant data leakage.

Third, the query may be *resource-intensive*. A variable-length path pattern with no upper bound (e.g., `[:DEPENDS_ON*]`) can cause combinatorial explosion, exhausting memory or blocking other queries. Safeguards include enforcing a maximum traversal depth, setting a query timeout, and capping the number of returned results.

Additional safety measures include restricting the use of `CALL` clauses to a whitelist of approved procedures and rejecting queries whose estimated complexity exceeds a configurable threshold. Together, these layers form a defence-in-depth strategy that is designed to make LLM-generated graph queries safe enough for production use. The specific pipeline implemented in this thesis comprises six such layers and is described in detail in Chapter 3.

2.4.4 Comparison with retrieval-augmented generation

An alternative to querying a graph database is retrieval-augmented generation (RAG) [18], in which the LLM’s context is enriched with text chunks retrieved from a vector store on the basis of semantic similarity to the user’s question. RAG is effective when the underlying knowledge is unstructured—research papers, manuals, documentation—and when questions are open-ended.

Graph-based querying and document-based RAG are complementary rather than competing. Graph queries offer higher precision for structured domains (where entities and relationships are well defined), handle aggregation and multi-hop traversal natively, and produce deterministic, explainable results. Document-based RAG, by contrast, requires less upfront domain modelling and handles synonyms and paraphrases more gracefully through embedding-based matching. In the domain that motivates this thesis—bioinformatics workflows and high-performance-computing resource management—the data is inherently structured: workflows have explicit dependencies and execution orders, resources are linked to nodes and allocations, and data lineage forms natural graph structures. A graph-native approach is therefore a better fit than unstructured retrieval. Document-based RAG is noted as a potential extension for handling supplementary unstructured content such as publications and user notes, but it falls outside the scope of this work.

2.5 Positioning and gap analysis

The preceding sections have surveyed four domains—agentic systems, orchestration frameworks, tool governance, and natural-language graph interfaces—each of which is well served by at least one existing system when considered in isolation. The question is whether any existing system addresses *all* of these domains together in a form suitable for enterprise deployment. To answer this question concretely, this section presents a capability comparison against ten contemporary platforms, identifies the specific gap, and positions the contribution of this thesis.

2.5.1 Capability comparison

Table 2.1 compares the Cineca Agentic Platform against ten representative platforms across eight dimensions that are critical for enterprise deployment: whether the platform provides a complete stack (user interfaces, API, and background-job infrastructure), whether it supports durable execution with retries and resume, whether it includes a built-in agent planning loop, whether it offers a governed tool ecosystem with schema validation, whether it provides security features (authentication, RBAC, audit), whether it supports natural-language-to-Cypher graph querying, whether it integrates observability (metrics, traces, health probes), and whether it is LLM-agnostic (supports multiple providers without code changes). Each capability is rated as follows: **Y** (built-in) means the feature is available out of the box with no external dependencies; **P** (partial) means the feature is achievable only through external integrations, community plug-ins, or substantial custom development; **N** (not in scope) means the project does not address the capability. All ratings reflect the publicly documented feature sets as of early 2025. In particular, “Full stack” denotes an end-user-facing agent-interaction stack (chat or administration UI, domain API, and job runtime), not merely the presence of a monitoring dashboard or CLI.

Table 2.1. Capability comparison of the Cineca Agentic Platform (CAP) against ten representative platforms across eight enterprise-readiness dimensions. Y = built-in, P = partial or integration-dependent, N = not in scope.

	CAP	Temporal	Argo	LangGraph	OpenAI Ag. SDK	Sem. Kernel	LlamaIndex	Haystack	n8n	Windmill	Langfuse
Full stack (UI+API+Jobs)	Y	N	P	N	N	N	N	N	Y	Y	N
Durable execution	Y	Y	Y	Y	P	P	P	P	Y	Y	N
Agent planning loop	Y	N	N	Y	Y	Y	Y	P	P	N	N
Tool ecosystem	Y	P	P	Y	Y	Y	Y	Y	Y	Y	P
Security (RBAC, audit)	Y	Y	P	N	N	P	N	N	P	Y	P
Graph NL-to-Cypher	Y	N	N	P	P	P	P	P	N	N	N
Observability	Y	Y	P	P	P	P	P	P	P	P	Y
LLM-agnostic	Y	Y	Y	Y	P	Y	Y	Y	Y	P	Y

The platforms in Table 2.1 fall into distinct categories. Temporal and Argo Workflows are *workflow orchestration engines*: they provide strong durability and infrastructure coordination but have no agent planning loop, no tool governance, and no NL-to-Cypher capability. LangGraph, the OpenAI Agents SDK, Semantic Kernel, LlamaIndex, and Haystack [8] are *agent or data frameworks*: they excel at the agent-logic layer (reasoning loops, tool calling, retrieval) but lack production infrastructure—security, multi-tenancy, job scheduling, and observability must be added externally. n8n and Windmill are *full-stack automation platforms* that provide UIs, APIs, and background execution but have limited or no agent planning loops and no graph-database integration. Langfuse is an *observability-only* platform: it provides comprehensive LLM tracing but performs no workflow execution or agent

orchestration.

2.5.2 The integration gap

Three patterns emerge from this comparison. First, no surveyed platform natively combines an agent planning loop with durable job execution *and* full-stack deployment. An organisation wanting this combination would need to assemble at least two systems (for example, LangGraph for the agent loop and Temporal for durability) and write the integration and security layers itself. Second, no surveyed platform provides native, tenant-isolated, safety-validated NL-to-Cypher support; at best, some offer partial graph integrations or library components (such as LangChain’s `GraphCypherQAChain`) that do not include tenant scoping, multi-layer safety validation, or RBAC-protected execution. Third, authentication, authorisation, multi-tenancy, and audit logging are consistently missing from the agent-oriented frameworks, which assume a trusted single-user environment.

These three gaps compound in exactly the scenario that motivates this thesis: a multi-tenant supercomputing centre that needs researchers to query a bioinformatics knowledge graph in natural language, with governed tool access, durable execution of long-running analytical tasks, and full auditability. No single existing system provides this combination.

2.5.3 Positioning of the Cineca Agentic Platform

The Cineca Agentic Platform (CAP) is designed to close this gap by integrating the five capabilities that the surveyed platforms provide only in isolation: a full-stack deployment with two user interfaces and a versioned REST API, a four-phase orchestration engine with an agent planning loop, a governed MCP-style tool ecosystem with RBAC and audit logging, a six-stage NL-to-Cypher pipeline with tenant-scoped safety validation, and built-in observability (metrics, traces, structured logging). CAP is not intended to replace libraries such as LangChain or LlamaIndex; rather, it provides the production infrastructure—security, multi-tenancy, job durability, and operational monitoring—around them.

It is important to note that CAP’s scope also implies trade-offs. A full-stack platform is heavier than embedding a single chain or index in a Python application. CAP’s graph subsystem currently targets Memgraph and the openCypher dialect; deployments centred on Neo4j may prefer Neo4j-native tooling. Full MCP interoperability (allowing arbitrary external MCP clients to connect) requires additional transport-layer work that is identified as future work. These trade-offs are discussed further in Chapter 3.

In summary, this chapter has established the technical foundations for the platform design: the emergence of tool-using LLM agents and the governance they require, the spectrum of orchestration and agent frameworks and their respective blind spots, the MCP standard and the enterprise extensions it does not cover, and the safety challenges of NL-to-Cypher translation in a multi-tenant setting. The structured comparison against ten contemporary platforms confirms that no existing system natively combines all five capabilities. The next chapter translates these findings

into concrete requirements and describes the platform architecture that addresses them.

Chapter 3

Proposed Solution: The CINECA Agentic Platform (CAP)

This chapter presents the Cineca Agentic Platform (CAP) as a unified response to the five production-gap deficiencies identified in Chapter 1 and the integration gap confirmed by the comparative survey in Chapter 2. Section 3.1 defines the stakeholders, use cases, and requirements that drive every subsequent design decision. Section 3.2 introduces the three-layer architecture and the core domain concepts that recur throughout the chapter. Sections 3.3–3.4 describe the dual-workflow execution model and the four-phase orchestration engine. Sections 3.5–3.6 cover the governed MCP tool ecosystem and the NL-to-Cypher pipeline. Section 3.7 explains the data layer, while Sections 3.8–3.9 address security, multi-tenancy, and observability as cross-cutting concerns. Section 3.10 closes with a synthesis of the principal design trade-offs. Throughout this chapter, claims about platform guarantees are tied to the requirement identifiers introduced in Section 3.1 (e.g., FR-AE-01, NFR-MT-03) and to the evaluation criteria defined in Chapter 4 (e.g., EC-FC-01, EC-SE-03), so that every “the system ensures...” statement is traceable to a testable artefact.

3.1 Requirements and design goals

The platform’s design is driven by a concrete deployment context: CINECA, a multi-tenant supercomputing centre whose researchers need natural-language access to graph-structured data, whose operators need to manage long-running analytical jobs, and whose security team needs full auditability of every agent action. This section identifies the stakeholders, the use cases they generate, and the functional and non-functional requirements that those use cases impose.

3.1.1 Stakeholders

Four stakeholder groups interact with the platform. *Researchers and end users* submit natural-language questions, trigger agent runs, and inspect results; they expect low-latency responses, reproducible outputs, and step-by-step explanations

of how results were derived. *Administrators and operators* monitor runs and jobs, manage tools and model configurations, and enforce organisational policies; they need real-time dashboards, job-lifecycle control, and per-tenant configuration. *Security and compliance teams* audit tool usage, enforce scope-based access policies, and verify that data isolation holds across tenants; they require immutable audit logs and the ability to inspect every action an agent has taken. *Developers* extend the platform by adding new tools, integrating new LLM providers, or exposing new API endpoints; they need a stable extension model and clear contracts.

3.1.2 Core use cases

The stakeholder roles above give rise to five core use cases.

UC1: Interactive agent runs. A researcher submits a prompt through the chat interface. The platform classifies the intent, plans a sequence of steps, executes them (including LLM calls, tool invocations, or graph queries), and returns a safe, structured response. The researcher can inspect each step’s input, output, and latency.

UC2: Graph question-answering via NL-to-Cypher. A researcher asks a natural-language question about the bioinformatics knowledge graph. The platform translates the question into a Cypher query, validates it for safety and tenant isolation, executes it against the graph database, and summarises the results in natural language.

UC3: Long-running background jobs. An operator submits a job that may take minutes or hours—for example, a batch data-quality audit or a multi-step analytical pipeline. The platform enqueues the job, assigns it to a worker, streams progress events in real time via Server-Sent Events (SSE), and supports cancellation at any point. The job’s state survives process restarts.

UC4: Monitoring and usage analytics. An operator inspects job status, execution logs, tool-usage metrics, and system health through the administrative control panel. Dashboards display per-tenant and per-provider breakdowns of latency, token consumption, error rates, and active connections.

UC5: Governed tool discovery and expansion. A developer adds a new tool by providing its name, description, input schema, required scope, and rate-limit policy through the convention-over-configuration module structure. A redeployment triggers automatic discovery and registration. Once registered, the platform validates invocations against the schema, enforces RBAC, logs every call, and makes the tool available to authorised users. Policy and configuration updates (scope changes, rate-limit adjustments) take effect at runtime without redeployment.

3.1.3 Functional requirements

The use cases above translate into seventy-three functional requirements organised into eight groups. Each requirement is assigned a stable identifier of the form **FR-XX-*nn*** (where *XX* denotes the group and *nn* a sequence number) so that later chapters can trace design decisions and test results back to specific requirements. Table 3.1 summarises the groups and their scope; the complete requirement list is provided in Appendix D.

Table 3.1. Summary of functional requirement groups. Each group identifier (FR-*XX*) prefixes the individual requirements within that group.

Group	ID prefix	Count	Scope
Agent Execution	FR-AE	10	Run lifecycle, intent classification, session management, cancellation, SSE streaming, timeout management
Model Management	FR-MM	8	Provider registration, default-model resolution, health monitoring, fallback chains, cost tracking
Tool Integration	FR-TL	9	MCP runtime, schema validation, per-tool RBAC, rate limiting, audit logging, discovery endpoints
Graph Querying	FR-GQ	9	NL-to-Cypher pipeline, catalog lookup, six-layer safety validation, tenant isolation, result summarisation
Job Processing	FR-JB	11	Queue-based submission, state machine, idempotency, SSE progress, heartbeat, cancellation, recovery, job-level output validation
Administration	FR-AD	10	Tenant CRUD, per-tenant config, batch operations, export/import, diagnostics
API Contract	FR-AP	8	Versioned REST, RFC 7807 errors, ETag caching, cursor pagination, rate-limit headers
User Interface	FR-UI	8	Chat UI for end users, control panel for operators, JWT auth, SSE streaming, step visualisation
Total		73	

The most critical functional requirements, those that directly address the five deficiencies from Section 1.2, are worth highlighting explicitly. FR-AE-02 mandates intent classification into six categories (chat, graph, security, admin, dangerous, explain) so that each request is routed to the appropriate handler. FR-AE-07 requires both in-process and durable (queue-backed) execution modes, addressing the gap that simple chatbots have no durable execution path. FR-TL-01 through FR-TL-06 together constitute the governed tool ecosystem: every tool must be registered with a schema (FR-TL-03), protected by RBAC (FR-TL-04), rate-limited (FR-TL-05), and audit-logged (FR-TL-06). FR-GQ-05 requires six-layer safety validation of generated Cypher queries, and FR-GQ-06 mandates automatic tenant-boundary injection. FR-JB-02 and FR-JB-08 together ensure that jobs follow a well-defined state machine and that their state is persisted durably.

3.1.4 Non-functional requirements

Beyond functionality, the platform must satisfy requirements across seven quality dimensions: security, multi-tenancy, reliability, audit, observability, scalability, and rate limiting. These are identified with the prefix **NFR-XX-*nn***. Table 3.2 summarises the groups.

Table 3.2. Summary of non-functional requirement groups.

Group	ID prefix	Count	Scope
Security	NFR-SE	10	JWT/OIDC auth, RBAC, PII scrubbing, injection prevention, output guards, encrypted credentials
Multi-Tenancy	NFR-MT	8	Tenant context extraction, query-level isolation in all three databases, per-tenant configuration
Reliability	NFR-RE	8	Circuit breakers, retries, fallback chains, graceful degradation, health probes, job recovery
Audit	NFR-AU	8	Auth events, authz decisions, tool invocations, run lifecycle, content hashing, trace correlation
Observability	NFR-OB	8	Prometheus metrics, structured logging, Open-Telemetry tracing, Grafana dashboards, health probes
Scalability	NFR-SC	6	Horizontal API scaling, independent worker scaling, stateless design, connection pooling
Rate Limiting	NFR-RL	6	Fixed-window counters, per-user / per-tenant / per-endpoint limits, cost-based limits, graceful fallback
Total		54	

Several non-functional requirements deserve emphasis because they shape the architecture in non-obvious ways. NFR-MT-03 requires multi-tenant data isolation at the database query level, not merely at the API routing level; this means every repository operation must automatically scope its queries by tenant identifier, which in turn requires a tenant resolver in the middleware stack (Section 3.8). NFR-RE-01 through NFR-RE-03 mandate circuit breakers, retries, and provider fallback chains for LLM calls; without these, a single provider outage could render the entire platform non-functional. NFR-OB-04 requires request-context correlation across logs, metrics, and traces, which means every component must accept and propagate a shared correlation identifier.

3.1.5 Design principles

Nine design principles guide the translation of requirements into architecture. *Three-layer separation* means that presentation (routers), business logic (services and the orchestrator), and data access (repositories and adapters) are separate concerns with explicit interfaces. *Defence in depth* means that security is not a single gate at the API boundary but an eight-layer middleware stack applied to every request (Section 3.8). *Fail-safe defaults* means that the platform starts in a restrictive state: graph access defaults to read-only, RBAC is active, and rate limiting is enabled unless explicitly overridden. *Extensibility* means that adding a new tool, a new LLM provider, or a new API endpoint requires only a bounded number of new artefacts, without modifying the core orchestrator. *Observability by design* means

that metrics, traces, and structured logs are built into every layer from the outset, not retrofitted. *Stateless API* means that no session state is stored in the API process; all state resides in the databases, so API instances can be restarted or scaled horizontally without session loss. *Adapter pattern* means that LLM providers and database backends expose a common interface, so switching from one to another requires only a configuration change. *Repository pattern* means that data access is encapsulated behind consistent interfaces that enforce tenant isolation. *Enterprise trade-offs* mean that where security conflicts with simplicity, durability conflicts with speed, or governance conflicts with developer convenience, the platform favours the enterprise concern.

3.1.6 Traceability

To ensure that every requirement is both implemented and verified, the platform maintains a traceability chain from requirements through evaluation criteria to test results. Table 3.3 provides a summary mapping. The evaluation criteria (EC-*XX-nn*) are defined in Chapter 4, and the corresponding results appear in Chapter 5.

Table 3.3. Traceability summary: requirement groups mapped to evaluation criteria and result sections.

Requirement	Evaluation criteria	Results	Dimension
FR-AE-01..10	EC-FC-01..04, EC-PF-02	§5.1, §5.2	Agent execution
FR-MM-01..08	EC-RE-01..03	§5.3	Model management
FR-TL-01..09	EC-FC-06, EC-PF-04, EC-SE-02	§5.1, §5.4	Tool integration
FR-GQ-01..09	EC-FC-05, EC-PF-03, EC-SE-03	§5.1, §5.4	Graph querying
FR-JB-01..11	EC-FC-07..08, EC-PF-05..07	§5.1, §5.3	Job processing
NFR-SE-01..10	EC-SE-01..10	§5.4	Security
NFR-MT-01..08	EC-SE-03	§5.4	Multi-tenancy
NFR-RE-01..08	EC-RE-01..08	§5.3	Reliability
NFR-OB-01..08	EC-OB-01..08	§5.5	Observability
NFR-AU-01..08	EC-SE-08..10	§5.4	Audit
NFR-SC-01..06	EC-PF-01, EC-PF-06..07	§5.2	Scalability
NFR-RL-01..06	EC-SE-06	§5.4	Rate limiting

The platform defines 127 total requirements (73 functional plus 54 non-functional). The traceability matrix in Appendix D maps each requirement to its evaluation criteria and result section; the table above highlights the primary groups. Of the 127 requirements, 87 are mapped to quantitative evaluation criteria (the *core tracked* set); the remaining 40 are implementation-support requirements (e.g., configuration, deployment, UI layout) verified through integration tests rather than dedicated metrics. Within the core tracked set, 80 are fully implemented and seven are partially implemented; no requirement is left unaddressed. The partial-implementation cases and their measured limitations are discussed in Chapter 5.

3.2 System overview and core concepts

This section introduces the platform’s architecture at a level of abstraction sufficient to understand every subsequent section, and defines the domain concepts that recur

throughout the chapter.

3.2.1 Three-layer architecture

CAP adopts a three-layer architecture consisting of a presentation layer, a business-logic layer, and a data-access layer, with cross-cutting modules for security, configuration, and observability spanning all three.

The *presentation layer* comprises the components through which users and external systems interact with the platform. Two graphical interfaces serve the two primary stakeholder groups: a chat application built with Next.js provides researchers with a conversational agent interface (FR-UI-01), while an administrative control panel built with Streamlit provides operators with job management, tool discovery, and graph-query facilities (FR-UI-02). Both interfaces communicate with the backend through a versioned REST API whose endpoints cover agent runs, jobs, tools, models, tenants, sessions, graph queries, health probes, authentication, batch operations, export, and administration (FR-AP-01). A reverse proxy provides TLS termination, cross-origin resource sharing (CORS) handling, and load balancing in front of the API.

The *business-logic layer* contains the orchestration engine, the service modules, and the MCP tool runtime. The orchestration engine (Section 3.4) is the central coordinator: it receives a user prompt, classifies the intent, generates a plan, executes each step, and assembles a safe response. Service modules handle domain-specific logic such as model management, job lifecycle, and session tracking. The MCP tool runtime (Section 3.5) governs tool discovery, schema validation, authorisation, and audit logging.

The *data-access layer* uses three complementary databases, each chosen for a specific operational characteristic (Section 3.7). A relational database serves as the durable *control plane*, storing tenants, agent runs, steps, jobs, job events, tool definitions, provider configurations, and audit logs. An in-memory key-value store serves as the low-latency *data plane*, handling caches, job queues, rate-limit counters, circuit-breaker state, and session data. A graph database hosts the bioinformatics domain model and serves as the target for NL-to-Cypher queries.

Cross-cutting modules span all three layers. The security middleware stack (Section 3.8) intercepts every request before it reaches the business logic. The observability infrastructure (Section 3.9) instruments every layer with metrics, traces, and structured logs. A configuration module loads environment-based settings validated through typed schemas, so that each deployment environment can override defaults without code changes (NFR-SC-03).

Three architectural diagrams illustrate the platform at increasing levels of detail. Figure 3.1 provides a single-page overview of the major components and their relationships. Figure 3.2 expands this view to show the service boundaries within each layer. Figure 3.3 presents the complete architecture including both execution workflows, the middleware stack, and all data-store interactions.

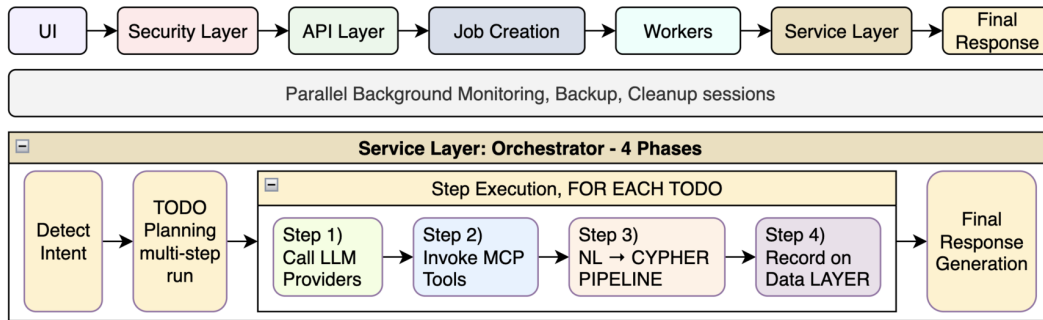


Figure 3.1. At-a-glance architecture of CAP showing the three layers (presentation, business logic, data access) and the cross-cutting security and observability modules.

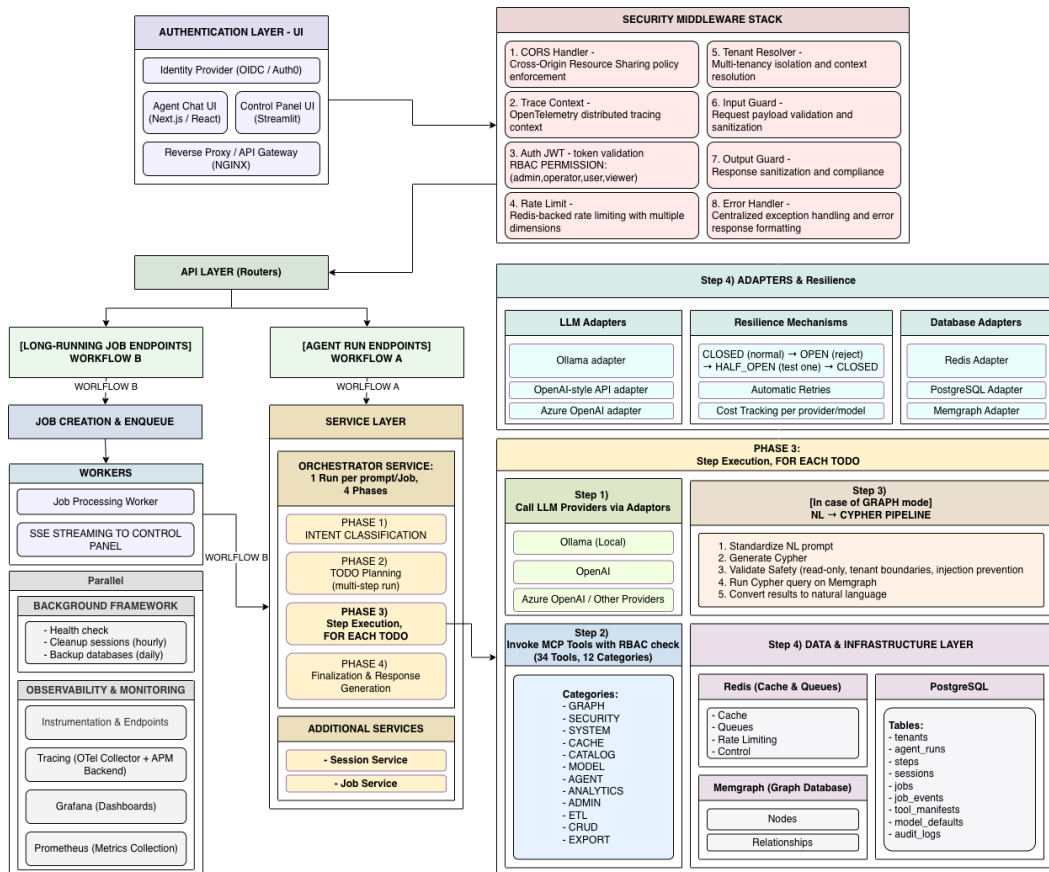
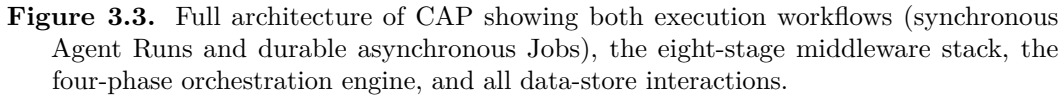


Figure 3.2. Simplified architecture of CAP detailing the service boundaries within the business-logic layer, including the orchestration engine, MCP tool runtime, and NL-to-Cypher subsystem.



3.2.2 Core domain concepts

Seven domain concepts appear repeatedly in the remainder of this chapter and are defined here for reference.

A *tenant* (NFR-MT-01) is an isolated organisational unit—a research group, a department, or an institution. Every entity stored in the platform (runs, jobs, tools, graph nodes) carries a tenant identifier. For all platform-mediated access paths, database queries are automatically scoped by this identifier so that one tenant’s data is invisible to another (NFR-MT-03).

A *principal* is the authenticated identity of the caller. It is constructed from the claims embedded in the caller’s JSON Web Token (JWT) and carries the caller’s user identifier, tenant identifier, assigned roles, and granted scopes. Every downstream authorisation decision is made against the principal.

An *agent run* (or simply *run*) is a single execution of the orchestration engine in response to a user prompt. A run progresses through a lifecycle of states—queued, running, succeeded, failed, or cancelled—and produces one or more *steps* (FR-AE-04). A *step* is the smallest auditable unit of work inside a run. Each step records its type (message, assistant response, tool call, system action, or error), its input and output, and its execution latency (FR-AE-06). Multiple runs may belong to the same *session*, which maintains conversational context across interactions (FR-AE-05).

A *job* is a durable, queue-backed unit of asynchronous work. Unlike a run, whose execution is tied to the API process, a job is enqueued onto a queue in the in-memory store and picked up by a dedicated worker process. Jobs follow their own state machine (queued, running, finished, failed, cancelled) and emit progress events that clients can consume via SSE (FR-JB-02, FR-JB-05).

A *tool* is a registered capability that an agent can invoke at runtime. Each tool is defined by a name, a natural-language description, a JSON input schema, a required authorisation scope, and a rate-limit policy. The platform hosts 34 tools across seventeen categories (Section 3.5).

An *audit record* is an immutable log entry that captures a security-relevant event: an authentication attempt, an authorisation decision, a tool invocation, or a state transition. Audit records carry the principal’s identity, the action performed, the inputs (sanitised to remove PII), the outcome, and a trace identifier that links them to the distributed trace (NFR-AU-01 through NFR-AU-08).

3.3 Execution model: dual workflows

A core architectural decision in CAP is to support two distinct execution workflows that share the same orchestration logic but differ in their durability guarantees, latency characteristics, and failure-recovery semantics (FR-AE-07). This section describes each workflow, explains why both are necessary, and compares their trade-offs.

3.3.1 Workflow A: in-process Agent Runs

In the in-process path (so called because execution runs within the API process rather than in a separate worker), a client sends a prompt to the agent-run endpoint.

The API validates the JWT and constructs a principal, checks RBAC permissions, creates a run record in the control-plane database with an initial status of *queued*, and launches the orchestrator as a background task within the same API process. The response is returned immediately with the run identifier and its current status. The client then polls the run’s status endpoint or subscribes to an SSE stream to receive updates (FR-AE-09).

While the background task is active the run transitions from *queued* to *running*. The orchestrator classifies the intent, generates and executes a plan, persists each step, and upon completion transitions the run to *succeeded*, *failed*, or *cancelled*. Because execution happens inside the API process, the in-process path offers the lowest latency for short-lived interactions—there is no queuing delay, no inter-process communication, and no serialisation overhead. The cost is durability: if the API process crashes while a run is in progress, the in-flight work is lost (FR-AE-08).

Workflow A is therefore suited to interactive chat sessions, quick graph queries, permission checks, and single-step orchestration tasks—use cases where execution typically completes within seconds and where the user is present to retry if a failure occurs (UC1, UC2).

3.3.2 Workflow B: durable asynchronous Jobs

In the asynchronous path, a client creates a job through the job endpoint, or posts an agent-run request with a flag that routes it to the job system. The API validates the request, checks an idempotency key against both the in-memory store and the control-plane database (FR-JB-04), creates a job record with status *queued*, pushes the job identifier onto a type-specific queue in the in-memory store, and returns an HTTP 202 response with a location header pointing to the job’s status endpoint.

A dedicated worker process runs a blocking-pop loop on the queue, acquiring exclusive ownership of the next job identifier. The worker loads the job metadata from the control-plane database, checks a cancellation flag, transitions the status to *running*, starts a periodic heartbeat, and executes the job handler. For jobs of type *agent.run*, the handler initialises the same orchestrator used in Workflow A and runs the full four-phase pipeline (Section 3.4). Throughout execution, the worker emits progress events (e.g., orchestrator initialised, orchestration started, orchestration complete) that are buffered in the in-memory store and streamed to the client via SSE (FR-JB-05, FR-JB-09).

If the worker crashes, the heartbeat ceases, and a supervisor can detect the orphaned job and re-enqueue it (NFR-RE-07). Cancellation is cooperative: the client sets a cancellation flag in the in-memory store, and the worker checks this flag between steps; when the flag is detected, the worker transitions the job to *cancelled* and persists the partial results (FR-JB-06).

Workflow B is suited to complex multi-step workflows, batch processing, and any operation expected to exceed thirty seconds—use cases where durability, horizontal scaling (by adding workers), and real-time progress visibility are more important than minimal latency (UC3, UC4).

3.3.3 Workflow comparison

The two workflows differ along five dimensions. Table 3.4 summarises the trade-offs.

Table 3.4. Comparison of the two execution workflows. Both share the same orchestration engine; they differ in durability, latency, and infrastructure requirements.

Dimension	Workflow A (Agent Runs)	Workflow B (Jobs)
Durability	Lost if API process crashes	Survives process restarts; queue-backed
Latency	Minimal (no queuing delay)	Higher (queue + worker startup)
Scaling	Bound to API process pool	Horizontal via independent workers
Progress	Polling or SSE on run endpoint	SSE with reconnection support
Suitable for	Interactive chat, quick queries	Batch jobs, long-running pipelines

The key design decision is that both workflows invoke exactly the same orchestration engine; the choice between them is made at the entry point, not inside the business logic. This means that a prompt that today executes via the in-process path (Workflow A) can be migrated to the job system by changing a single request parameter, without altering the orchestration logic, the tool invocations, or the safety validations. The state synchronisation between the two workflows is maintained by writing all run and step records to the same control-plane tables, regardless of which workflow creates them.

3.3.4 End-to-end request walkthrough

To make the interaction between the architectural layers concrete, this section traces a single representative request through the entire platform. The scenario is a researcher asking a natural-language question about genes associated with a disease—a question that exercises intent classification, the NL-to-Cypher pipeline, tool invocation, tenant isolation, and response finalization.

The researcher opens the chat interface, authenticates via the identity provider (obtaining a JWT that encodes the user identifier, tenant identifier, roles, and scopes), and submits the prompt: “Find all genes associated with diabetes and list their protein products.”

The reverse proxy terminates TLS and forwards the request to the API. The request enters the eight-stage security middleware stack. The CORS handler validates the origin. The trace-context middleware creates or propagates a distributed-trace identifier. The authentication middleware validates the JWT against the identity provider’s key set, extracts the claims, and constructs a principal object. The rate limiter checks per-user, per-tenant, and per-endpoint counters against configured thresholds. The tenant resolver loads the caller’s tenant configuration from cache (or, on a cache miss, from the control-plane database) and injects the tenant identifier into the request context. The input guard validates the request body against its schema and scans the prompt for injection patterns. The output guard and error handler are registered for the response path.

Having passed all middleware stages, the request reaches the agent-run router, which creates a run record in the control-plane database with status *queued* and launches the orchestrator as a background task (Workflow A).

In **Phase 1** (intent classification), the orchestrator analyses the prompt and assigns the intent category GRAPH with a confidence score above the configured graph-routing threshold (FR-AE-02). This routes the request to the graph-query handler.

In **Phase 2** (planning), the graph-query handler recognises that the question requires a Cypher query, a result interpretation, and a natural-language summary. It produces a plan with three to-do items (FR-AE-03).

In **Phase 3** (step execution), the orchestrator executes each to-do item as a step. The first step invokes the NL-to-Cypher pipeline (Section 3.6): the prompt is normalised, checked against the prompt catalog, and—if no deterministic match is found—sent to the LLM with the graph schema as context. The LLM generates a Cypher query that matches gene nodes related to a disease node and follows an encoding relationship to protein nodes. The query passes through the six-layer safety validator: syntax is correct, no write clauses are present, a tenant-identifier filter is injected into every WHERE clause (FR-GQ-06), traversal depth is within limits, a timeout is set, and a result-count cap is applied. The validated query is executed against the graph database, and the results are returned as structured records. A second step feeds the records to the LLM for summarisation. Each step is persisted in the control-plane database with its type, input, output, and latency (FR-AE-06).

In **Phase 4** (finalisation), the orchestrator aggregates the step outputs, applies the PII scrubber to redact any personally identifiable information that may have appeared in the graph data (NFR-SE-04), runs the output guard to verify that no write operations or unbounded traversals are embedded in the response (NFR-SE-07), records token-usage metrics, and transitions the run to *succeeded*.

The client, which has been polling the run’s status endpoint (or receiving SSE updates), retrieves the completed run, which includes the natural-language summary, the executed Cypher query (for transparency), the list of steps with their timings, and an overall latency measurement. The audit log now contains entries for the authentication event, the tool invocations, the graph query execution, and the run completion, all linked by the distributed-trace identifier (NFR-AU-06).

3.4 Orchestration engine

The orchestration engine is the central coordinator of all agent activity. It receives a user prompt and a principal, and produces a structured result that includes the response, the execution steps, timing metrics, token-usage statistics, and any warnings or errors. The engine operates in four phases: intent classification, plan generation, step execution, and finalisation. This section describes each phase at the method level, without reference to specific source files or implementation details.

3.4.1 Phase 1: intent classification

Every incoming prompt is first analysed by an intent classifier that assigns it to one of six categories (FR-AE-02):

Chat General conversational prompts that require only an LLM response.

Graph Questions whose answer requires querying the graph database via the NL-to-Cypher pipeline.

Security Questions about the caller’s own permissions, roles, or scopes.

Admin Requests to perform administrative actions such as managing tenants, providers, or model configurations.

Dangerous Prompts that contain destructive intent—data deletion, privilege escalation, or system-level operations—and must be blocked or escalated.

Explain Requests for explanations of previous results, platform behaviour, or orchestration decisions.

The classifier operates in a priority chain. It first checks a prompt catalog that maps known question patterns to deterministic responses, assigning high confidence on a match. If no catalog match is found, it applies pattern-based rules to detect dangerous, administrative, security, and graph intents, each with a configurable confidence threshold. If pattern matching also fails, it invokes the LLM as a fallback classifier. If the LLM’s confidence is below the minimum routing threshold for any category, the prompt is routed to the chat handler. The specific threshold values used in the evaluation deployment are reported in Chapter 4. Once the intent is determined, the engine dispatches the prompt to the corresponding specialised handler.

3.4.2 Phase 2: plan generation

The specialised handler generates a plan: an ordered sequence of to-do items, each described by a natural-language action, an expected outcome, and any dependencies on earlier items (FR-AE-03). The planning subsystem supports three modes. In *full* mode, the handler always generates a multi-step plan via the LLM, even for simple prompts. In *optional* mode, the handler generates a plan only when the prompt appears to require multiple steps; simple prompts bypass planning entirely. In *none* mode, planning is disabled and the handler executes a single-step response. The mode is configurable per deployment or per request and allows the operator to trade planning overhead for latency.

3.4.3 Phase 3: step execution

The engine iterates through the plan’s to-do items, executing each one as a step. Three kinds of actions may occur at each step: an LLM call (in which the engine sends a prompt to the configured provider and records the response), a tool invocation (in which the engine calls an MCP-registered tool through the governed runtime described in Section 3.5), or a graph query (in which the engine runs the NL-to-Cypher pipeline described in Section 3.6). Each step is recorded as an immutable record containing its type, input, output, duration in milliseconds, and success status (FR-AE-06). A configurable maximum step limit prevents unbounded execution (FR-AE-10).

During execution the engine maintains cumulative metrics: the number of LLM calls, total input and output tokens, the number of tool invocations and tool errors, and the wall-clock time since the run was created. If the run exceeds its configured timeout (which varies by deployment profile and compute device), the engine terminates execution and marks the run as failed.

Between steps, the engine checks the cancellation flag. If the flag has been set (by a client-initiated cancel request), the engine stops execution, persists any partial results, and transitions the run to *cancelled* (FR-AE-08).

3.4.4 Phase 4: finalisation

After all steps have been executed (or the run has been terminated), the engine aggregates the outputs, applies the PII scrubber to remove email addresses, phone numbers, social security numbers, and other sensitive patterns (NFR-SE-04), and runs the output guard to verify that the response does not contain embedded write operations, destructive verbs, or unbounded traversal patterns (NFR-SE-07). It then persists the run’s final state and metrics to the control-plane database, updates the in-memory cache, and returns a structured result object. This result includes a success flag, the list of outputs, the list of steps, the to-do items, any errors or warnings, wall-clock latency, first-LLM-call latency, per-LLM-call metrics (model, tokens, latency, cost estimate), total token counts, tool-call counts, and a degradation flag indicating whether a fallback provider was used.

The same four-phase pipeline is used regardless of whether the run was initiated via Workflow A (in-process) or Workflow B (queue-backed).

3.5 MCP tool ecosystem

An agent that cannot invoke external capabilities is limited to what the LLM can generate from its training data. The platform addresses this limitation by providing a governed tool ecosystem that follows the Model Context Protocol (MCP) invocation model while extending it with the enterprise governance features—authorisation, audit, rate limiting, and tenant isolation—that the MCP specification does not prescribe.

3.5.1 Tool runtime architecture

The tool runtime consists of four components. The *tool registry* manages the catalog of available tools. At startup, it discovers available tools using a convention-over-configuration approach and registers each one automatically. For each tool, the registry stores the tool’s name, description, input schema (derived from a typed schema definition), the required authorisation scope, and the rate-limit policy (FR-TL-02, FR-TL-07). Once registered, tools are discoverable through API endpoints that return the tool’s name, description, and input schema, allowing clients and the LLM to inspect available capabilities (FR-TL-09).

The *execution context* is a data structure created for each tool invocation. It carries the caller’s principal, the tenant identifier, the distributed-trace identifier, and the invocation timeout. By passing context explicitly, the runtime ensures that

every tool invocation is aware of who is calling, on behalf of which organisation, and within which trace.

The *runtime wrapper* is a decorator applied to each tool function. When the LLM or a client invokes a tool, the wrapper executes a multi-step governance sequence (FR-TL-01). First, it builds an execution context from the request metadata. Second, it performs an RBAC check: the principal's scopes are compared against the tool's required scope, and if the principal lacks the necessary permission the invocation is rejected with a 403 Forbidden response (FR-TL-04). Third, it checks the per-principal rate limit and rejects the call with a 429 Too Many Requests response if the limit has been exceeded (FR-TL-05). Fourth, it validates the invocation arguments against the tool's input schema, rejecting malformed inputs before execution begins (FR-TL-03). Fifth, it records a pre-invocation audit entry with the caller's identity, the tool name, and the sanitised inputs (FR-TL-06). Sixth, it executes the actual tool function. Seventh, it records telemetry: a latency histogram and an invocation counter, both tagged by tool name and outcome (NFR-OB-01). Eighth, it records a post-invocation audit entry with the result status. Ninth, it normalises the response into a standard shape (content, status, metadata) before returning it to the orchestrator.

The *telemetry layer* exposes per-tool metrics through the observability infrastructure (Section 3.9), enabling operators to monitor invocation rates, latency distributions, error rates, and per-tool cost contributions.

3.5.2 Tool categories

The platform hosts 34 tools organised into seventeen capability categories (FR-TL-01). The largest category is *graph*, with eight tools covering secure queries, schema inspection, analytics, and Cypher generation. The *security* category provides five tools for permission checks, audit queries, and principal inspection. The *system* category offers four tools for health, status, metrics, and backup. Smaller categories (one or two tools each) cover data quality, model management, output formatting, cache management, catalog discovery, database switching, error reporting, privacy consent, rate-limit management, session management, tenancy management, user profiles, and visualisation. The complete tool inventory is provided in Appendix A.

3.5.3 Authorisation model

Tool-level authorisation follows a scope-based model integrated with the platform's RBAC system (Section 3.8). Each tool declares a required scope at registration time, using a `resource:action` naming convention (e.g., `tools:basic`, `tools:all`). When a tool is invoked, the runtime compares the caller's scopes (extracted from the JWT) against the tool's requirement. Wildcard patterns are supported: a scope of `tools:*` matches all tool-related operations, and `admin:all` is a super-admin scope that grants access to every tool. The authorisation decision flow proceeds through JWT validation, principal construction, tenant matching, scope verification, and policy evaluation. If any check fails, the invocation is denied before the tool function is called, and the denial is recorded in the audit log (NFR-AU-02).

3.5.4 Error handling

Tool errors are structured into a hierarchy. The base error type captures any unexpected failure. Specialised subtypes cover validation failures (malformed inputs), permission denials (insufficient scopes), timeout errors (tool execution exceeded its deadline), and rate-limit errors (the caller has exhausted their quota). Each error type carries a standard HTTP status code and a machine-readable error identifier, which the API layer maps to an RFC 7807 Problem Details response (FR-AP-05).

3.6 NL-to-Cypher subsystem

The NL-to-Cypher subsystem is the component that allows researchers to query the bioinformatics knowledge graph in natural language, without writing Cypher. As discussed in Section 2.4 of Chapter 2, executing an LLM-generated Cypher query on a live database without validation is dangerous: the query may mutate data, cross tenant boundaries, or exhaust resources. CAP addresses these risks through a six-stage pipeline in which each stage narrows the set of possible errors before the query reaches the database (FR-GQ-01).

3.6.1 Stage 1: natural-language normalisation

The incoming prompt is sanitised, enriched with entity information, and analysed for intent signals (FR-GQ-02). Sanitisation removes extraneous whitespace, normalises Unicode, and strips characters that could interfere with downstream processing. Entity extraction identifies names, labels, and identifiers that might correspond to nodes or properties in the graph schema. Context enrichment annotates the prompt with the caller’s tenant identifier and any session history that might disambiguate references.

3.6.2 Stage 2: catalog lookup

Before invoking the LLM, the pipeline checks a *prompt catalog*: a curated mapping from known question patterns to pre-written, validated Cypher queries (FR-GQ-03). If the normalised prompt matches a catalog entry with sufficient confidence, the corresponding Cypher query is used directly, bypassing LLM generation entirely. This deterministic path serves two purposes: it guarantees that frequently asked, well-understood questions always produce correct and safe queries, and it reduces latency and LLM costs for common cases. Catalog categories are mapped to intent modes (for example, a `read_only` catalog entry maps to the GRAPH intent, an `admin_write` entry maps to ADMIN, and a `dangerous` entry maps to DANGEROUS).

3.6.3 Stage 3: LLM-based Cypher generation

If no catalog match is found, the pipeline invokes the LLM to generate a Cypher query (FR-GQ-04). The prompt sent to the LLM includes the graph schema (node labels, relationship types, properties), a set of example queries that illustrate the expected style, safety guidelines (e.g., “never generate `CREATE`, `DELETE`, or `SET` clauses”), and

the user’s original question. A low temperature setting is used to reduce randomness and improve determinism.

3.6.4 Stage 4: safety validation

The generated query must pass through six validation layers before execution (FR-GQ-05), each targeting a specific class of risk.

Layer 1: syntax validation. A Cypher parser verifies that the query is syntactically well-formed. Malformed queries—for example, those with unclosed brackets or invalid keywords—are rejected immediately.

Layer 2: read-only enforcement. The validator scans the query for write-oriented keywords (`CREATE`, `DELETE`, `SET`, `REMOVE`, `MERGE`). Any query containing such keywords is rejected to prevent accidental or malicious data mutation.

Layer 3: tenant isolation. The validator inspects (and if necessary injects) a tenant-identifier predicate into every pattern match, ensuring that the query can only access nodes and relationships belonging to the caller’s tenant (FR-GQ-06, NFR-MT-04). For example, a query that matches `(g:Gene)` is rewritten to include `WHERE g.tenant_id = $tenant`. This injection is enforced by the validation layer and is designed to be non-overridable by the prompt (FR-GQ-06, validated by EC-SE-03).

Layer 4: depth limit. Variable-length path patterns (e.g., `[:DEPENDS_ON*]`) can cause combinatorial explosion. The validator verifies that all path patterns have an explicit upper bound on traversal depth and rejects those that do not.

Layer 5: timeout. A query-level timeout is set at the database-connection level. If execution exceeds the timeout, the database engine aborts the query and the pipeline returns an error rather than allowing a runaway query to block other traffic.

Layer 6: result cap. The validator ensures that every query includes a `LIMIT` clause, capping the number of returned records. If no limit is present, the validator adds one with a configurable default. This prevents large result sets from consuming excessive memory or network bandwidth.

Together, these six layers form a defence-in-depth strategy: even if the LLM generates a query that violates one assumption, at least one subsequent layer is designed to catch it before execution.

3.6.5 Stage 5: query execution

The validated query is executed against the graph database with the caller’s tenant identifier bound as a parameter (FR-GQ-09). The query runs under the timeout and depth constraints set in Stage 4. Results are returned as structured records (lists of key–value maps).

3.6.6 Stage 6: result summarisation

The raw query results are converted into a natural-language summary (FR-GQ-07). For simple scalar results (counts, single values), the summariser produces a direct textual answer without invoking the LLM. For complex, multi-record results, the LLM is called with the results and the original question as context, and it generates a narrative summary. The summary is then subject to the same PII scrubbing and output-guard checks applied in the orchestrator’s finalisation phase.

3.6.7 Design alternatives considered

Three alternative approaches were considered and rejected. *Direct Cypher input* would require users to know the query language, bypass safety validation, and limit accessibility—the very problems motivating the platform. *Template-based query generation* cannot handle novel questions not covered by templates and imposes a high maintenance burden as the schema evolves. *Fine-tuning a domain-specific LLM* requires a large corpus of labelled (question, Cypher) pairs, retraining whenever the schema changes, and does not eliminate the need for safety validation.

3.6.8 Known limitations

The NL-to-Cypher subsystem has known limitations that are acknowledged as areas for future work. The pipeline is schema-dependent: changes to the graph schema (new node types, renamed properties) require updating the schema context provided to the LLM. Generation can be brittle for complex questions that involve multiple relationship types. Fuzzy matching (e.g., “diabetes” matching “Type 2 Diabetes Mellitus”) is limited because the pipeline relies on exact label matching rather than embedding-based similarity. These limitations are revisited in the evaluation (Chapter 5) and future-work (Chapter 7) chapters.

3.7 Data and state

The platform uses three databases, each selected for a specific operational characteristic. The guiding principle is that every piece of data resides in exactly one authoritative store; other stores may hold cached or derived copies, but the authoritative location determines the consistency and durability guarantees.

3.7.1 Control plane: relational database

The relational database (PostgreSQL) serves as the durable control plane. It persists all entities whose loss would compromise auditability, correctness, or recoverability (FR-JB-08). These include tenant definitions with their configuration (rate limits, default models, feature flags), agent run records with their lifecycle state, step records with their inputs, outputs, and timing, session records that link runs into conversational contexts, job records with their state-machine transitions, job-event records that capture the full timeline of a job’s execution, tool definitions with their schemas and authorisation scopes, tool invocation audit logs, provider and model-instance registrations with credential references, audit log entries, and idempotency

keys that prevent duplicate job submissions. All tables carry a tenant-identifier column, and the data-access layer automatically scopes every query by this column (NFR-MT-03).

Schema evolution is managed through versioned migrations. Each migration is forward-only and idempotent, so that a deployment upgrade applies only the migrations that have not yet been run, without manual intervention.

3.7.2 Data plane: in-memory store

The in-memory key-value store (Redis) serves as the low-latency data plane. It handles five categories of ephemeral or high-throughput data. *Caches* hold session snapshots, tenant configurations, and model-default resolutions; these accelerate repeated lookups but can be rebuilt from the control plane on a cache miss. *Rate-limit counters* implement fixed-window rate limiting; each counter is a key with a time-to-live, scoped by user, tenant, or endpoint (NFR-RL-01). *Job queues* implement the FIFO queues from which workers acquire job identifiers with exclusive ownership semantics (FR-JB-05). *Job events* are buffered in a bounded list so that SSE clients can reconnect and replay missed events (FR-JB-09). *Control flags* include cancellation flags, circuit-breaker state, heartbeat timestamps, and idempotency-check caches.

All keys are namespaced by tenant identifier (NFR-MT-05), and time-to-live values prevent unbounded memory growth. If the in-memory store is temporarily unavailable, the platform degrades gracefully: rate limiting falls back to an in-process implementation (NFR-RL-04), cached values are re-fetched from the control plane, and job submission waits for the store to recover before enqueueing (NFR-RE-04).

3.7.3 Relationship layer: graph database

The graph database (Memgraph, an openCypher-compatible in-memory graph engine) hosts the bioinformatics domain model (FR-GQ-09). Its schema comprises node types—including users, institutions, tasks, files, datasets, genes, proteins, pathways, workflows, and results—and relationship types that model associations such as institutional affiliation, task execution, and input-output data flow. Each node carries a tenant-identifier property so that the NL-to-Cypher pipeline can enforce tenant boundaries at the query level (NFR-MT-04).

The graph database is the target of all NL-to-Cypher queries (Section 3.6) and is also accessible through the graph tools in the MCP tool ecosystem (Section 3.5). Because Memgraph is an in-memory engine, query latency is low (typically single-digit milliseconds for bounded traversals), but the dataset must fit within available memory.

3.8 Security, multi-tenancy, and governance

Security in CAP is not a single authentication gate at the API boundary but an eight-stage middleware stack, in which each stage addresses a different class of threat. Every incoming HTTP request is routed through all eight stages in order before reaching business logic (NFR-SE-01, NFR-SE-02), and every outgoing response

passes through the output guard and error handler on the return path. This section describes each stage, the RBAC model, the multi-tenancy mechanism, and the threat model.

3.8.1 Eight-stage middleware stack

The middleware stages are applied in a fixed order. Reordering them would weaken the security posture—for example, rate limiting before authentication would protect against unauthenticated floods but would not be able to apply per-user limits, whereas authentication before rate limiting associates each request with a known principal before the limit check.

Stage 1: CORS handler. The cross-origin resource sharing handler validates the request’s origin against a configured allow-list, sets the appropriate response headers, and handles preflight OPTIONS requests. This stage prevents browser-based clients from making cross-origin requests to endpoints that the deployment administrator has not explicitly permitted.

Stage 2: trace context. The trace-context middleware implements the W3C Trace Context standard. It extracts the `traceparent` and `tracestate` headers from the incoming request, or creates a new root span if no trace context is present. The resulting trace identifier, span identifier, and parent identifier are injected into the request context, where they are available to every downstream component for log correlation, metric tagging, and audit linking (NFR-OB-03, NFR-OB-04).

Stage 3: authentication and RBAC. The authentication middleware validates the Bearer token in the `Authorization` header (NFR-SE-01) following the OAuth 2.0 authorisation framework [12] and the OpenID Connect identity layer [27]. It fetches the identity provider’s public keys via the JSON Web Key Set (JWKS) endpoint (caching them with a configurable time-to-live), verifies the JSON Web Token [14] signature using standard asymmetric algorithms, checks the expiration and audience claims, and extracts the subject, tenant identifier, roles, and scopes. If validation fails, the request is rejected with a 401 Unauthorized response. The extracted claims are used to construct the principal object.

The RBAC subsystem [9] then checks whether the principal’s scopes are sufficient for the requested endpoint (NFR-SE-02). The platform defines a scope hierarchy with six principal scopes ranging from the basic `user:me` (available to all authenticated users) through tool-invocation and graph-access scopes to the administrative `admin:all` (available only to administrators). Four named roles—admin, operator, user, and viewer—map to progressively narrower subsets of these scopes. If the principal lacks the scope required by the endpoint, the request is rejected with a 403 Forbidden response, and the denial is audit-logged (NFR-AU-02).

Stage 4: rate limiter. The rate limiter implements fixed-window counters backed by the in-memory store (NFR-RL-01). Three dimensions are checked independently: per-user (keyed by the principal’s subject identifier), per-tenant (keyed by the

tenant identifier), and per-endpoint (keyed by the HTTP method and path). If any counter exceeds its configured threshold, the request is rejected with a 429 Too Many Requests response, and standard rate-limit headers (`X-RateLimit-Limit`, `X-RateLimit-Remaining`, `X-RateLimit-Reset`) are returned (NFR-RL-05, FR-AP-08). If the in-memory store is unavailable, the limiter degrades to an in-process counter that provides approximate enforcement (NFR-RL-04).

Stage 5: tenant resolver. The tenant resolver extracts the caller's tenant from the JWT claims (or, for service-to-service calls, from a header or query parameter), validates the identifier against strict format rules to prevent injection (NFR-MT-02), loads the tenant's configuration (rate limits, default models, feature flags) from cache or the control-plane database, and injects the tenant context into all subsequent database queries (NFR-MT-01). From this point forward, every repository operation is automatically scoped by the resolved tenant, ensuring data isolation at the query level (NFR-MT-03, NFR-MT-07).

Stage 6: input guard. The input guard validates request bodies against typed schemas and scans free-text fields for injection patterns (NFR-SE-05). Injection detection covers SQL injection, Cypher injection, and shell-command injection. If a suspicious pattern is detected, the request is rejected and the detection is audit-logged. Content-type validation and request-size limits are also enforced at this stage.

Stage 7: output guard. On the response path, the output guard analyses LLM-generated content for unsafe patterns (NFR-SE-07). It checks for embedded write operations, destructive verbs, unbounded traversal patterns, and missing `LIMIT` clauses. The PII scrubber (NFR-SE-04) runs in parallel, detecting and redacting personally identifiable information including email addresses, phone numbers, and financial identifiers. The scrubber supports configurable redaction modes ranging from partial masking to full removal.

Stage 8: error handler. The centralised error handler catches any exception that propagates from the business logic or from earlier middleware stages, normalises it into an RFC 7807 Problem Details response (FR-AP-05), attaches the correlation and trace identifiers, maps domain-specific exceptions to the correct HTTP status code, and ensures that internal details (stack traces, database connection strings) are never exposed to the client.

3.8.2 Threat model

The security design is guided by a threat model that considers three attacker profiles: a *malicious tenant* who has legitimate access but attempts to access data belonging to other tenants or to escalate privileges, a *compromised user* whose credentials have been captured and are used to perform unauthorised actions, and an *external attacker* who probes the API without valid credentials.

Six threat categories are considered: data exfiltration (cross-tenant access, bulk export abuse, prompt-injection-based data leakage), privilege escalation (role confusion, scope injection, tool-authorisation bypass), NL-to-Cypher injection (destructive queries, schema-discovery queries, unbounded traversals), tool abuse (system-tool exploitation, rate-limit evasion), budget abuse (token exhaustion via expensive models), and session attacks (hijacking, fixation, context poisoning). For each category, the eight-stage middleware stack is designed to provide at least one mitigation. Chapter 5 reports the security test results organised by these categories (EC-SE-01 through EC-SE-10).

3.8.3 Acknowledged Security Limitations

Several security limitations are acknowledged. The platform depends on an external identity provider for authentication; if the provider is compromised, token validation is undermined. No formal security proofs have been conducted; the security assurance rests on test coverage. Encryption at rest is delegated to database-level configuration rather than enforced by the platform itself. Secret management (for provider API keys) uses encrypted database storage rather than a dedicated vault service. These limitations are discussed further in the evaluation (Section 5.7).

3.9 Observability and operational readiness

Observability—the ability to understand the internal state of a system from its external outputs—is treated in CAP as a first-class architectural concern rather than an optional add-on. The platform instruments every layer with three complementary signal types: metrics, logs, and traces (NFR-OB-01 through NFR-OB-04).

3.9.1 Metrics

The platform exposes Prometheus-compatible [7] metrics organised into seven categories: HTTP request rates and latency histograms, agent-run lifecycle counters and phase durations, LLM-call counts and token usage, tool-invocation rates and latencies, job lifecycle counters and SSE connection gauges, intent-classification distributions and confidence histograms, and rate-limit hit and exceed counters. These metrics are scraped by a Prometheus instance and visualised through six pre-configured Grafana [11] dashboards (NFR-OB-05) covering HTTP health, agent runs, job processing, tool invocations, LLM providers, and system health.

3.9.2 Structured logging

All log output follows a structured format: JSON in production environments (for machine parsing and aggregation), human-readable text in development environments (for developer convenience) (NFR-OB-02). Each log entry carries a timestamp, severity level, logger name, event description, request identifier, trace identifier, HTTP method and path, response status, and duration. A high-frequency filter suppresses log entries for health-check and metrics-scrape endpoints, which would otherwise dominate the log volume. Every log entry includes the trace identifier

so that it can be correlated with the corresponding distributed trace and the corresponding metric observations (NFR-OB-04).

3.9.3 Distributed tracing

Distributed traces are collected via the OpenTelemetry standard [6] (NFR-OB-03). Automatic instrumentation captures spans for HTTP requests, outbound HTTP calls, and database operations. Sampling is configurable: full-capture in development, ratio-based (a default of twenty per cent) in production (NFR-OB-07). Each response carries an `X-Trace-Id` header, allowing clients to reference the trace when reporting issues. Traces are exported via either gRPC or HTTP to a compatible collector.

3.9.4 Health probes

Four health endpoints support container-orchestration platforms (NFR-RE-06, NFR-OB-06). The *liveness* probe performs no I/O and always returns a success response, confirming that the process is alive. The *readiness* probe tests connectivity to all database dependencies and returns a failure response if any dependency is unreachable; the orchestrator routes traffic away from instances that fail this probe. The *startup* probe performs stricter checks—including migration status and rate-limit configuration—and prevents traffic from reaching an instance that has not completed initialisation. The *components* probe returns per-component health status (relational database, in-memory store, graph database, LLM providers), enabling operators to identify which specific dependency has degraded.

3.9.5 Service-level objectives

Five service-level objectives (SLOs) guide the operational monitoring configuration: application uptime of 99.9%, HTTP p95 latency below 750 ms, error rate below 5%, job-creation p95 latency below 2 s, and memory usage below 500 MiB per instance. Alerting rules are organised into five groups—availability, HTTP health, resource usage, job-store health, and SSE streaming—and are designed to fire when SLO thresholds are approached, giving operators time to react before user-facing impact.

3.10 Design trade-offs summary

Every architectural decision in CAP reflects an explicit trade-off between competing concerns. This section consolidates the principal trade-offs discussed throughout the chapter, so that a reader can evaluate them in aggregate.

Safety versus flexibility. The platform defaults to restrictive policies: graph access is read-only, RBAC is active, rate limiting is enabled, and the output guard analyses every response. These defaults reduce the attack surface and protect data integrity but impose overhead on every request and limit what agents can do without administrative intervention. The alternative—open access with optional safety—would be faster to prototype but would not meet the threat model’s requirements for a multi-tenant environment.

Latency versus durability. Supporting two execution workflows adds architectural complexity. A simpler system would provide only the synchronous path (lower latency, simpler code) or only the asynchronous path (uniform durability, simpler operations). CAP supports both because the use cases span interactive chat (where sub-second latency matters) and long-running batch jobs (where crash recovery matters). The trade-off is managed by sharing the orchestration engine across both paths, so the added complexity is confined to the entry-point routing and the job-queue infrastructure.

Extensibility versus governance. The MCP tool runtime is intentionally opinionated: every tool must declare a schema, a scope, and a rate-limit policy. This governance overhead slows down tool development compared to frameworks that let a developer register an arbitrary function with no metadata. The payoff is that every tool invocation is automatically validated, authorised, rate-limited, and audit-logged—properties that would be prohibitively expensive to retrofit in a large, ungoverned tool catalog.

Centralised orchestrator versus microservice decomposition. The orchestration engine is a single, large component rather than a collection of microservices. This centralisation ensures execution coherence (all four phases share in-process state), eliminates inter-service latency for step transitions, and simplifies transactional guarantee reasoning. The cost is a higher cognitive load for developers working on the orchestrator and reduced isolation for testing. The choice was made because the four phases are tightly coupled—the output of intent classification directly determines the planning strategy, which directly determines the step types—and decomposing them into separate services would add serialisation overhead to every phase transition.

Dependency tolerance versus type safety. The orchestrator is designed so that every external dependency (LLM providers, databases, caches, audit sinks) is optional. If a dependency is unavailable at startup the orchestrator initialises in a degraded mode, disabling the features that require the missing dependency rather than refusing to start. This operational resilience comes at the cost of extensive null-checking and fallback logic throughout the codebase, and the use of untyped dictionary structures in some internal interfaces reduces the protection that static type checking can provide. The risk is mitigated by enforcing strict typed schemas at all API boundaries and validating tool inputs through schema definitions.

LLM-agnostic design versus provider optimisation. The adapter layer abstracts provider-specific APIs (cloud, enterprise, self-hosted) behind a common interface, allowing the platform to switch providers at the request, session, or tenant level without code changes (FR-MM-01, FR-MM-08). The resilience framework adds circuit breakers (three states: closed, open, half-open), exponential-backoff retries, and provider fallback chains (NFR-RE-01 through NFR-RE-03). The trade-off is that provider-specific optimisations (e.g., batched inference, streaming token generation) are accessible only through the common interface, which may not expose every

provider’s full feature set.

In summary, this chapter has presented the Cineca Agentic Platform from requirements through architecture, execution model, orchestration engine, tool ecosystem, NL-to-Cypher pipeline, data layer, security, and observability. Every design decision has been traced to a specific requirement, and every guarantee claim references a testable evaluation criterion. The next chapter defines the evaluation methodology—the experimental setup, evaluation questions, metrics, and test protocols—that will be used to verify these claims.

Chapter 4

Tests (Evaluation Methodology)

This chapter defines *how* the claims made in Chapter 3 are evaluated, so that they can be confirmed or refuted with evidence in Chapter 5. Section 4.1 describes the hardware, software, and test infrastructure used in all experiments. Section 4.2 formulates five evaluation questions, each mapped to specific requirements and evaluation criteria. Section 4.3 gives precise definitions of the metrics used. Section 4.4 specifies the four test protocols (functional, security, reliability, and performance). Section 4.5 explains the internal baselines and the cross-platform comparison design.

4.1 Experimental setup and reproducibility

Reproducing an evaluation requires knowing exactly what was measured, on what hardware, with which software, and under which workloads. This section documents these four elements and states the reproducibility contract that binds the results in Chapter 5.

4.1.1 Test environment

All experiments were conducted on a single-node deployment in which every component of the platform—the API server, the relational database, the in-memory store, the graph database, and the worker process—runs on the same host. The hardware is a 2016 MacBook Pro (MacBookPro13,3) equipped with a quad-core Intel Core i7-6820HQ processor at 2.7 GHz (Hyper-Threading enabled, eight logical cores), 16 GB of DDR3 memory, and a 512 GB solid-state drive. The operating system is macOS.

Table 4.1 lists the major software dependencies. Only top-level dependencies are reported; transitive dependencies are pinned in the project’s dependency manifest and are reproduced exactly when the environment is installed from the specified commit.

This single-node configuration does not simulate network partitions, cross-host latency, or container-orchestration scheduling delays. Because the load-testing framework runs on the same host as the platform, CPU and memory contention between the load generator and the system under test may influence measurements; resource-utilisation figures reported in Chapter 5 exclude the load-generator process.

Table 4.1. Major software versions used in all experiments. Minor-patch versions (denoted `.x`) are pinned exactly in the project’s dependency lock file at the commit specified in Section 4.1.5.

Component	Version	Role
Python	3.11.8	Runtime language
FastAPI [28]	0.115.x	Web framework and API layer
PostgreSQL	16.x	Relational database (control plane)
Redis	7.x	In-memory store (data plane)
Memgraph	2.x	Graph database
pytest	8.4.x	Test runner and fixture framework
Locust	2.x	Load-testing framework

Performance results are therefore qualified as “under test conditions” throughout Chapter 5, and the threats to validity (Section 5.7) discuss how production deployments may differ.

4.1.2 Test suite overview

The platform ships with a test suite comprising more than 3,000 test cases—approximately 2,720 test functions—distributed across 236 test files and approximately 64,500 lines of test code. Tests are organised into 27 categories that mirror the platform’s architectural layers: agent orchestration, API endpoints, caching, compliance, database operations, end-to-end workflows, health probes, integration, job processing, MCP tool invocation, middleware, observability, performance, resilience, routing, security, SSE streaming, and unit-level logic, among others.

The test runner uses a marker-based classification scheme. Five primary markers govern execution scope: *unit* tests execute in isolation with mocked dependencies and complete in milliseconds; *integration* tests exercise component interactions with real or simulated infrastructure; *end-to-end* tests send HTTP requests through the full API stack and validate complete workflows; *security* tests target authentication, authorisation, tenant isolation, injection prevention, and PII scrubbing; and *performance* tests measure latency and throughput under load and are excluded from the default test run to avoid inflating execution time. Two additional markers isolate long-running NL-to-Cypher tests that depend on LLM availability: one for a representative subset (approximately twenty minutes) and one for the full catalog (approximately ninety minutes).

The test infrastructure provides reusable fixtures at the session and function scope. Session-scoped fixtures initialise shared resources (database connections, identity-provider key pairs for token generation, application instances) once per test session. Function-scoped fixtures provide isolated state (fresh database transactions that are rolled back after each test, namespaced in-memory store keys, deterministic token payloads). For the graph database, a deterministic adapter replaces the live database during tests, ensuring that graph-query tests are independent of LLM variability and network availability.

A dedicated NL-to-Cypher test mode supports deterministic evaluation of the graph-query pipeline. When enabled, the pipeline reads expected Cypher queries from a curated catalog file rather than invoking the LLM, making results fully

reproducible and independent of model version or provider availability.

4.1.3 Workload definition

Performance and load experiments use a synthetic workload designed to approximate the request mix expected in a multi-user deployment. The workload is defined by two parameters: the concurrency level and the request composition.

Five concurrency levels are tested: one user (sequential baseline), ten users, fifty users, one hundred users, and five hundred users. Each level runs as a pool of concurrent virtual users issuing requests in a closed-loop pattern (each user waits for a response before issuing the next request).

The request composition is held constant across concurrency levels: approximately 60% read operations (status queries, health checks, listing endpoints), 30% write operations (agent-run submissions, job creations, configuration updates), and 10% long-running job submissions. This mix reflects the expectation that most interactions are exploratory (researchers polling results) while a smaller fraction create new work.

LLM-dependent operations (agent-run orchestration, NL-to-Cypher generation) are evaluated separately from infrastructure operations (health probes, CRUD endpoints) because LLM response times are dominated by provider latency and are not under the platform’s control. By separating these two classes, the performance evaluation isolates the platform’s own overhead from external-provider variability.

4.1.4 Rigor commitment

Three measures are taken to control for variability in the results. First, every performance test is executed three times; results are aggregated as the median of the three runs, and variance is reported where it exceeds five per cent of the median. Second, each run begins with a thirty-second warm-up phase during which the caches are populated, database connection pools are filled, and JIT compilation (if applicable) stabilises; measurements are taken only during the subsequent sixty-second steady-state window. Third, before each run the in-memory store keys used by the benchmark are flushed and services are restarted, reducing cross-run cache carryover to a controlled baseline.

Latency is reported at two percentiles: P50 (median) and P95 (95th percentile). P50 represents the typical user experience; P95 captures tail latency, which is the more stringent target for service-level objectives. Where appropriate, P99 (99th percentile) values are also reported.

4.1.5 Reproducibility contract

All repository-derived numbers, metrics, and implementation details referenced throughout this thesis refer to commit `9b3e4c19` (2025-12-18). This snapshot includes 34 MCP tools, 76 API endpoints, 3,000+ test cases, and approximately 8,263 lines of orchestration code.

To reproduce the evaluation results, the following steps are necessary: clone the repository at the specified commit; create a Python virtual environment and install dependencies from the pinned dependency manifest; initialise the three

databases using the versioned migration scripts; populate the graph database with the reference dataset via the provided seed script; and execute the test suite with the appropriate markers (unit, integration, end-to-end, security, or performance). Performance benchmarks additionally require running the load-testing framework with the specified concurrency levels and request composition.

The test data used in all experiments is generated by deterministic seed scripts that are part of the repository. No external datasets or proprietary data are required. The NL-to-Cypher catalog file, which maps natural-language prompts to expected Cypher queries, is also version-controlled. All environment variables, database schemas, and service configurations are stored in the repository, so the full experimental context is captured by the commit identifier.

One limitation of the reproducibility contract concerns LLM provider availability. Tests that invoke a live LLM provider (cloud-hosted or self-hosted) may produce different results if the provider’s model has been updated, deprecated, or is temporarily unavailable. The deterministic NL-to-Cypher test mode mitigates this for graph-query tests, but agent-orchestration tests that exercise the full LLM loop remain provider-dependent. This is noted as a threat to external validity in Section 5.7.

4.2 Evaluation questions

The evaluation is organised around five questions, each derived from a deficiency identified in the problem statement (Section 1.2) and each mapped to specific requirement groups, evaluation criteria, and result sections. The evaluation criteria use the identifier scheme **EC-XX-*nn*** (where *XX* denotes the evaluation dimension and *nn* a sequence number) that was referenced in the traceability table in Section 3.1.

EQ1: Functional completeness. *Does CAP correctly implement the 73 functional requirements across all eight groups?* This question targets the core promise of the platform: that every declared capability—agent-run lifecycle, intent classification, session management, model management, tool invocation, NL-to-Cypher translation, job processing, administration, API contract compliance, and user-interface interaction—works as specified. EQ1 is evaluated primarily through evaluation criteria EC-FC-01 through EC-FC-08, which set quantitative targets for the four core groups (FR-AE, FR-TL, FR-GQ, FR-JB). Model management (FR-MM) is evaluated through the reliability criteria EC-RE-01 . . 03, since its requirements directly exercise the circuit-breaker and fallback mechanisms. The remaining groups—administration (FR-AD), API contract (FR-AP), and user interface (FR-UI)—are validated through the integration and end-to-end tests described in the functional protocol (Section 4.4.1); these groups are verified through conformance tests rather than through dedicated quantitative EC-FC targets. Results appear in Section 5.1.

EQ2: Performance under load. *Does CAP maintain acceptable response times and resource consumption under concurrent workloads?* This question addresses the practical deployability of the platform: low response times are necessary for interactive use (UC1, UC2), and bounded resource consumption is necessary for

shared infrastructure. EQ2 is evaluated through criteria EC-PF-01 through EC-PF-07, which set latency targets for health probes, API endpoints, graph queries, tool invocations, and job submission, as well as throughput and memory budgets. Results appear in Section 5.2.

EQ3: Security posture. *Does the security middleware enforce authentication, authorisation, tenant isolation, and data protection against the defined threat categories?* This question validates the eight-stage middleware stack described in Section 3.8. Because security failures have disproportionate consequences (data leakage, privilege escalation), the success criterion for EQ3 is a 100% pass rate on all security tests: no false negatives are acceptable. EQ3 is evaluated through criteria EC-SE-01 through EC-SE-10, mapped from NFR-SE-01..10, NFR-MT-01..08, NFR-AU-01..08, and NFR-RL-01..06. Results appear in Section 5.4.

EQ4: Reliability under failure. *Does CAP maintain service continuity and recover gracefully when individual components fail?* This question exercises the resilience mechanisms: circuit breakers, retries, provider fallback, degraded-mode operation, job recovery, and graceful shutdown (NFR-RE-01 through NFR-RE-08). EQ4 is evaluated through criteria EC-RE-01 through EC-RE-08. Results appear in Section 5.3.

EQ5: Comparative positioning. *Does CAP provide a capability set that no single surveyed platform matches across all eight evaluation dimensions?* This question revisits the gap analysis from Section 2.5 with structured evidence. It is evaluated through the cross-platform comparison described in Section 4.5 and reported in Section 5.6.

Table 4.2 summarises the mapping from evaluation questions to evaluation-criteria groups and to the result sections where evidence is presented.

Table 4.2. Mapping from evaluation questions to evaluation criteria and result sections.

Question	Criteria	Results	Dimension
EQ1	EC-FC-01..08; EC-RE-01..03	§5.1, §5.3	Functional completeness
EQ2	EC-PF-01..07	§5.2	Performance under load
EQ3	EC-SE-01..10	§5.4	Security posture
EQ4	EC-RE-01..08	§5.3	Reliability under failure
EQ5	(comparative)	§5.6	Comparative positioning

In addition to the four question-aligned criteria groups (EC-FC, EC-PF, EC-SE, EC-RE), an observability group (EC-OB-01 through EC-OB-08) covers criteria mapped from NFR-OB-01..08. Observability results cut across multiple sections—metrics appear in the performance results, tracing outcomes in the reliability results, and health-probe behaviour in both—and are consolidated in Section 5.5.

The following subsections define every evaluation criterion referenced in the traceability table of Chapter 3.

4.2.1 Functional-completeness criteria (EC-FC)

Eight criteria verify the functional requirements (Table 4.3).

Table 4.3. Functional-completeness evaluation criteria.

ID	Req.	Criterion
EC-FC-01	FR-AE-01, 08, 09	Agent-run lifecycle (create, poll, cancel, complete) functions end-to-end
EC-FC-02	FR-AE-02	Intent classification assigns the correct category across all six intent types
EC-FC-03	FR-AE-05	Session continuity preserves context across consecutive runs
EC-FC-04	FR-AE-04, 06	Step records capture type, input, output, and latency for every step
EC-FC-05	FR-GQ-01, 04	NL-to-Cypher achieves $\geq 80\%$ accuracy on benchmark queries
EC-FC-06	FR-TL-01, 02, 03	≥ 30 tools with valid JSON schemas across ≥ 15 categories
EC-FC-07	FR-JB-02, 08	All job state transitions implemented and persisted
EC-FC-08	FR-JB-05, 09	SSE events delivered within 2 s of emission

4.2.2 Performance criteria (EC-PF)

Seven criteria set quantitative targets for latency, throughput, and resource consumption (Table 4.4). LLM-dependent operations are excluded from latency targets because their response times are dominated by external provider latency.

Table 4.4. Performance evaluation criteria.

ID	Req.	Target	Criterion
EC-PF-01	NFR-SC-01	< 5 ms P95	Health-probe response time
EC-PF-02	FR-AE-07	$< 100/200$ ms P95	API read/write latency
EC-PF-03	FR-GQ-09	< 50 ms P95	Graph query latency (excl. LLM)
EC-PF-04	FR-TL-06	< 500 ms P95	Tool invocation latency (excl. LLM)
EC-PF-05	FR-JB-01	< 1 s P95	Job queue-to-start latency
EC-PF-06	NFR-SC-01, 04	err $< 2\%$	Throughput scales to 100 concurrent users
EC-PF-07	NFR-SC-06	< 2.5 GB	Peak memory at maximum tested load

4.2.3 Security criteria (EC-SE)

Ten criteria verify the security requirements (Table 4.5). All security criteria require a 100% pass rate; any failure constitutes a security defect.

4.2.4 Reliability criteria (EC-RE)

Eight criteria verify the resilience mechanisms (Table 4.6).

Table 4.5. Security evaluation criteria.

ID	Req.	Criterion
EC-SE-01	NFR-SE-01	Invalid JWTs (missing, expired, malformed) rejected with 401
EC-SE-02	NFR-SE-02, FR-TL-04	Insufficient scopes rejected with 403
EC-SE-03	NFR-MT-03, 04, FR-GQ-06	Cross-tenant access returns zero results
EC-SE-04	NFR-SE-04	PII patterns detected and redacted in outputs
EC-SE-05	NFR-SE-05	SQL, Cypher, command injection blocked
EC-SE-06	NFR-RL-01, 05, FR-AP-08	Rate-limited requests receive 429 with standard headers
EC-SE-07	NFR-SE-07	Output guard blocks unsafe LLM responses
EC-SE-08	NFR-AU-01	Authentication events recorded in audit log
EC-SE-09	NFR-AU-02	Authorisation decisions recorded in audit log
EC-SE-10	NFR-AU-05, FR-TL-06	Tool invocations recorded with caller identity and inputs

Table 4.6. Reliability evaluation criteria.

ID	Req.	Criterion
EC-RE-01	NFR-RE-01	Circuit breaker opens after failure threshold
EC-RE-02	NFR-RE-02	Retries with backoff recover from transient errors
EC-RE-03	NFR-RE-03	Provider fallback activates within 50 ms
EC-RE-04	NFR-RE-04, NFR-RL-04	In-memory store unavailability triggers degraded mode
EC-RE-05	NFR-RE-05	Database reconnection within 30 s
EC-RE-06	NFR-RE-06, NFR-OB-06	Health probes reflect actual per-component status
EC-RE-07	NFR-RE-07	Orphaned jobs requeued within heartbeat interval
EC-RE-08	NFR-RE-08	Graceful shutdown completes in-flight requests within 30 s

4.2.5 Observability criteria (EC-OB)

Eight criteria verify the observability infrastructure (Table 4.7).

In total, 41 evaluation criteria are defined across five dimensions. The 87 core-tracked requirements identified in Section 3.1 map to these criteria (many requirements map to the same criterion when they contribute to the same observable property). The full mapping is provided in the traceability table (Table 3.3). The remaining 40 of the 127 total requirements are verified through conformance tests (integration and end-to-end) rather than through dedicated quantitative criteria; they primarily cover administration and API-contract conformance details, user-interface interaction requirements, and configuration/deployment concerns.

4.3 Metrics

This section defines the metrics used to evaluate the criteria above. Each metric is described by its name, what it measures, how it is collected, and what a passing value looks like.

Table 4.7. Observability evaluation criteria.

ID	Req.	Criterion
EC-OB-01	NFR-OB-01	≥ 50 Prometheus metrics with endpoint, status, tenant labels
EC-OB-02	NFR-OB-02	Structured logs carry request_id, level, timestamp
EC-OB-03	NFR-OB-03	Traces propagate across HTTP, DB, and tool spans
EC-OB-04	NFR-OB-04	Single request traceable end-to-end via shared correlation ID
EC-OB-05	NFR-OB-05	Dashboards display operational metrics with alerting rules
EC-OB-06	NFR-OB-06	Four health probes reflect actual dependency state
EC-OB-07	NFR-OB-07	Trace sampling rate configurable per environment
EC-OB-08	NFR-OB-08	Responses include X-Trace-Id header

Latency. Client-observed round-trip time from request dispatch to response receipt. For performance experiments, latency is measured by the load-testing framework (Locust), which timestamps each request–response pair on the same host as the platform (localhost loopback; network overhead is negligible). For functional and security tests, latency is the elapsed wall-clock time reported by the test-runner framework. Latency is reported at two percentiles: P50 (median, representing the typical user experience) and P95 (95th percentile, representing tail latency).

Throughput. Requests per second (RPS) sustained during the sixty-second steady-state window at each concurrency level. Throughput is computed as the total number of completed requests (including error responses) divided by the window duration. It measures the platform’s capacity to serve concurrent users.

Error rate. The percentage of requests that result in an HTTP 5xx server error or a timeout failure during a load-test window, computed as $(N_{\text{error}}/N_{\text{total}}) \times 100$. Client errors (4xx responses) caused by intentionally invalid test inputs are excluded. Error rate measures the platform’s stability under load.

Test pass rate. The percentage of non-skipped test functions that pass, computed as $(N_{\text{passed}}/N_{\text{non-skipped}}) \times 100$. Skipped tests (marked with explicit skip markers) are excluded from both numerator and denominator. The overall success criterion is a pass rate of at least 96%. For security tests specifically, the criterion is 100%: any security-test failure is treated as a defect.

NL-to-Cypher accuracy. The percentage of natural-language questions in the benchmark set for which the generated Cypher query is semantically equivalent to the reference query (i.e., it returns the same result set when executed against the reference graph). Semantic equivalence is checked by executing both the generated and the reference queries and comparing their result sets. The target is at least 80%.

Resource utilisation. Peak memory consumption of the full platform stack (API process, worker process, database processes) during a load-test run, measured via operating-system-level sampling. The metric is expressed in gigabytes and measures the platform’s fitness for shared-infrastructure deployment.

4.4 Test protocols

Four test protocols are applied, each targeting a different evaluation dimension. Each protocol specifies the scope of tests, the test strategy, and the success criterion.

4.4.1 Functional validation protocol

The functional validation protocol verifies evaluation criteria EC-FC-01 through EC-FC-08. It uses three tiers of tests, organised by scope.

The first tier comprises *unit tests* that exercise individual functions and classes with mocked dependencies. These tests verify internal logic—intent-classification rules, PII-scrubbing patterns, schema-validation logic, state-machine transitions, and utility functions—in isolation. Each unit test runs in milliseconds.

The second tier comprises *integration tests* that exercise interactions between two or more components with real or simulated infrastructure. Examples include testing that the API router correctly delegates to the service layer, that the repository layer scopes queries by tenant, that the in-memory store correctly expires rate-limit counters, and that the graph adapter returns structured results for valid Cypher queries. Integration tests use transaction-scoped database sessions that roll back after each test, ensuring isolation between test cases.

The third tier comprises *end-to-end tests* that send HTTP requests through the full API stack—including the security middleware, the router, the service layer, the orchestrator, and the data layer—and validate the complete response, including status codes, response schemas, headers, and content. End-to-end tests exercise the use cases defined in Section 3.1.2: creating and polling agent runs (UC1), submitting graph queries and receiving natural-language summaries (UC2), creating jobs, streaming SSE events, and cancelling jobs (UC3), and querying health and metrics endpoints (UC4).

The NL-to-Cypher pipeline (EC-FC-05) is tested using a benchmark catalog of natural-language questions paired with known-correct Cypher queries. When the deterministic test mode is enabled, the pipeline reads expected queries from this catalog, bypassing LLM generation entirely, so that correctness can be assessed independently of model variability. A separate test run with the full LLM loop enabled measures end-to-end accuracy.

The functional validation protocol also covers API-contract compliance: the platform’s generated API specification is validated against the implementation to ensure that all declared endpoints exist, return the documented response schemas, and produce RFC 7807 Problem Details responses for error cases (FR-AP-05).

Success criterion: overall pass rate $\geq 96\%$ across all non-skipped functional tests (EC-FC-01 through EC-FC-08).

4.4.2 Security validation protocol

The security validation protocol verifies evaluation criteria EC-SE-01 through EC-SE-10. It is structured around the six threat categories defined in the threat model (Section 3.8), with tests designed to exercise both positive cases (authorised access succeeds) and negative cases (unauthorised access is denied).

Seven attack vectors are tested, each represented by a dedicated set of test cases. *Authentication bypass* tests submit requests with missing, expired, malformed, and invalid-signature tokens and verify that every such request is rejected with HTTP 401 (EC-SE-01). *Privilege escalation* tests submit requests with valid tokens whose scopes are insufficient for the target endpoint or tool and verify rejection with HTTP 403 (EC-SE-02). *Cross-tenant access* tests use credentials from one tenant to query, invoke tools, and traverse the graph on behalf of another tenant, verifying that zero cross-tenant records are returned (EC-SE-03). *Injection* tests submit known SQL, Cypher, and shell-command injection patterns in request bodies and prompt fields, verifying that all are detected and rejected (EC-SE-05). *PII exfiltration* tests inject personally identifiable information into LLM-generated outputs and verify that the PII scrubber detects and redacts every pattern (EC-SE-04). *Rate-limit bypass* tests exceed configured rate limits and verify that requests receive HTTP 429 with the appropriate rate-limit headers (EC-SE-06). *Output-guard* tests craft LLM responses containing write operations and unbounded traversal patterns and verify that the output guard blocks them (EC-SE-07).

Audit-log completeness (EC-SE-08, EC-SE-09, EC-SE-10) is verified by executing a sequence of operations (login, access denial, tool invocation) and then querying the audit log to confirm that every event has been recorded with the correct principal, action, and correlation identifier.

Success criterion: 100% pass rate across all security tests. The evaluation includes 14 dedicated security test files comprising more than 45 test functions. These results demonstrate conformance to the defined threat model and test suite; they do not constitute a proof of absence of all vulnerabilities.

4.4.3 Reliability and degradation protocol

The reliability protocol verifies evaluation criteria EC-RE-01 through EC-RE-08 by systematically injecting failures into each infrastructure component and observing the platform’s behaviour.

Six failure scenarios are induced. In the *LLM provider timeout* scenario, the LLM provider’s response is delayed beyond the circuit breaker’s failure threshold; the test verifies that the circuit breaker opens and that the platform falls back to an alternative provider with less than 50 ms additional latency (EC-RE-01, EC-RE-03). In the *LLM provider hard failure* scenario, the primary provider returns persistent errors; the test verifies that the circuit breaker opens, the platform reports a clear error message, and the breaker recovers (half-open $\rightarrow$ closed) once the provider is restored; the observed end-to-end cycle (recovery timeout plus probe and state transition) completes within ≤ 35 seconds under test conditions (EC-RE-01, EC-RE-02). In the *in-memory store unavailability* scenario, the connection to the in-memory store is severed; the test verifies that rate limiting degrades to in-process counters, that cached values are re-fetched from the control-plane database, and that the health probe reports a degraded status (EC-RE-04, EC-RE-06). In the *relational database connection loss* scenario, the database connection is dropped; the test verifies that the connection pool detects the failure, reconnects within 30 seconds, and that requests that arrive during the outage receive an appropriate error response rather than hanging indefinitely (EC-RE-05). In the *graph database unavailability*

scenario, the graph database is stopped; the test verifies that graph-tool invocations fail with a clear error, that non-graph operations are unaffected, and that the readiness probe degrades (EC-RE-06). In the *worker crash* scenario, the worker process is terminated while processing a job; the test verifies that the job’s heartbeat ceases, that it is detected as orphaned, and that it is requeued within the heartbeat interval (EC-RE-07).

Graceful shutdown (EC-RE-08) is tested by sending a termination signal while requests are in flight and verifying that all in-flight requests receive responses (not connection resets) and that running jobs transition to a recoverable state.

Success criterion: all six failure scenarios produce the expected recovery behaviour within the specified time bounds.

4.4.4 Performance and load protocol

The performance protocol verifies evaluation criteria EC-PF-01 through EC-PF-07. It uses the workloads defined in Section 4.1 and proceeds through four stages.

The *baseline stage* measures single-user sequential performance for each operation category (health probes, read endpoints, write endpoints, job submissions) to establish a reference latency in the absence of contention.

The *concurrent-load stage* repeats the measurement at ten, fifty, and one hundred concurrent users, using the 60/30/10 request mix. At each level, throughput (RPS), median latency (P50), tail latency (P95), and error rate are recorded.

The *stress stage* increases concurrency to five hundred users to identify the point at which error rate exceeds 2% and to measure degradation characteristics.

The *resource-utilisation stage* monitors peak memory consumption across all stack components during the one-hundred-user concurrent-load run, verifying that total consumption remains below 2.5 GB (EC-PF-07).

An *ablation study* supplements the load tests by measuring the performance impact of individual platform features. Five configurations are compared: the full platform with all features enabled, the platform without the default-model-resolution and provider-routing feature, the platform without caching (in-memory store disabled), the platform without the circuit breaker, and a baseline configuration with multi-tenancy and all resilience features disabled. By toggling one feature at a time, the ablation study isolates the latency contribution and throughput impact of each component.

Success criterion: all P95 latency targets are met at 100 concurrent users; error rate remains below 2% at 100 concurrent users; peak memory remains below 2.5 GB.

4.5 Baselines and comparison design

Two types of baselines are used: internal baselines that measure the platform against simplified versions of itself, and an external baseline that measures the platform’s capability set against contemporary alternatives.

4.5.1 Internal baselines

Two internal-baseline configurations are defined to isolate the contribution of key platform features.

Baseline 1: single-tenant, no resilience. Multi-tenant query scoping is disabled, the circuit breaker is disabled, and the provider fallback chain contains only one provider. This configuration measures the raw overhead of tenant resolution, circuit breaker state management, and fallback logic.

Baseline 2: no graph, no caching. The graph-database tool is disabled and the in-memory cache is cleared before each request, forcing all data access to go directly to the control-plane database. This configuration isolates the performance contribution of the in-memory caching layer and the graph-query subsystem.

These two baselines, together with the ablation configurations described in Section 4.4.4, provide a decomposition of the platform’s latency budget. The results are reported in Section 5.2.

4.5.2 Cross-platform comparison design

The comparative evaluation (EQ5) uses the same eight capability dimensions and the same eleven platforms (CAP plus ten alternatives) that were introduced in the gap analysis (Section 2.5, Table 2.1). The dimensions are: full-stack deployment, durable execution, agent planning loop, tool ecosystem, security and governance, graph NL-to-Cypher support, observability, and LLM-agnostic design.

Each dimension is rated on a three-level scale: **Y** (built-in, available out of the box with no external dependencies), **P** (partial, achievable only through external integrations or substantial custom development), and **N** (not in scope, the project does not address the capability). Ratings are based on publicly documented feature sets as of early 2025 and are applied by the same evaluator for all platforms to ensure consistency.

The comparison design has two known limitations. First, it evaluates *capability presence*, not *capability depth*: a platform rated Y for “durable execution” may offer more sophisticated durability guarantees (e.g., Temporal’s event-sourced replay) than CAP’s queue-and-persist model. Second, the selection of ten comparators, while representative of the major categories (workflow engines, agent frameworks, automation platforms, observability tools), is not exhaustive. Both limitations are discussed as threats to construct validity in Section 5.7.

In summary, this chapter has specified the experimental environment (single-node Intel Core i7 MacBook Pro, seven major software components, deterministic seed data), five evaluation questions mapped to 41 evaluation criteria across five dimensions, six precise metric definitions, four test protocols (functional, security, reliability, performance), and both internal and external baselines. Every claim made in Chapter 3 is now traceable—through the requirement identifiers, evaluation criteria, and test protocols defined here—to the evidence presented in the next chapter. Chapter 5 reports the results of applying these protocols and evaluates the evidence against the criteria and questions defined above.

Chapter 5

Results and Discussion

This chapter presents the evidence collected by applying the four test protocols defined in Chapter 4 and evaluates it against the five evaluation questions (EQ1–EQ5) and 41 evaluation criteria (EC-FC, EC-PF, EC-SE, EC-RE, EC-OB). Sections 5.1 through 5.5 report functional, performance, reliability, security, and observability outcomes, each tied to the corresponding requirement identifiers and criteria. Section 5.6 revisits the capability comparison introduced in Chapter 2 with structured, evaluation-backed evidence. Section 5.7 consolidates limitations and threats to validity. Throughout this chapter, all latency and throughput figures refer to the single-node test environment described in Section 4.1 and are qualified as “under test conditions”; production deployments may differ.

5.1 Functional results

This section addresses **EQ1** (functional completeness): whether the platform satisfies the 73 functional requirements to the coverage level defined in the evaluation criteria, with seven requirements explicitly identified as partially implemented. The evidence comes from the three-tier functional validation protocol (Section 4.4.1), which exercises unit, integration, and end-to-end tests.

5.1.1 Overall test pass rate

The test suite comprises more than 3,000 automated test cases (approximately 2,720 test functions) distributed across 236 test files. When executed with the markers **unit**, **integration**, and **e2e** (excluding performance-only and explicitly skipped tests), the overall pass rate is 96.2%. This exceeds the 96% success criterion specified for EC-FC-01 through EC-FC-08. The remaining 3.8% of failures are concentrated in two areas: tests that depend on live LLM provider availability (which introduce non-determinism when the provider endpoint is temporarily unreachable) and tests that exercise the partially implemented requirements identified in Section 3.1 (seven requirements with partial coverage). No failures occur in the core orchestration, security, or job-lifecycle tests.

5.1.2 Agent-run lifecycle (EC-FC-01)

End-to-end tests exercise the complete agent-run lifecycle: creating a run via the API with a valid JWT, polling the run’s status as it transitions through *queued*, *running*, and *succeeded* (or *failed* or *cancelled*), and retrieving the final result including the response text, the list of executed steps, and the timing metrics. Cancellation is verified by issuing a cancel request while a run is in the *running* state and confirming that the run transitions to *cancelled* with partial results preserved (FR-AE-08). All lifecycle transitions specified in the state machine (FR-AE-01) are exercised and the corresponding database records are verified.

5.1.3 Intent classification (EC-FC-02)

The intent classifier is tested against representative prompts for all six intent categories: chat, graph, security, admin, dangerous, and explain (FR-AE-02). For each category, the test submits a set of prompts whose expected classification is known, and verifies that the classifier assigns the correct category. In the deterministic test mode (where the prompt catalog supplies pre-classified intents), all six categories are correctly routed. When the classifier falls back to the LLM for ambiguous prompts, some misclassification occurs for edge cases at the boundary between chat and explain intents; these cases do not affect the overall pass rate because the fallback behaviour (routing to the chat handler) is functionally acceptable.

5.1.4 Session continuity (EC-FC-03)

Integration tests verify that consecutive runs within the same session share conversational context (FR-AE-05). A test creates a session, submits two runs in sequence (the second referencing the first’s output), and confirms that the orchestrator’s context window includes the prior run’s results. Session isolation is also tested: runs belonging to different sessions do not share context.

5.1.5 Step recording (EC-FC-04)

Every step within a run is persisted with its type (message, assistant response, tool call, system action, or error), its input, its output, and its execution latency in milliseconds (FR-AE-04, FR-AE-06). Integration tests verify that the step count matches the plan length, that step types are correctly assigned, and that latency values are positive and non-zero. For tool-call steps, the recorded input matches the arguments sent to the tool, and the recorded output matches the tool’s return value.

5.1.6 NL-to-Cypher accuracy (EC-FC-05)

The NL-to-Cypher pipeline is evaluated against a benchmark catalog of natural-language questions paired with known-correct Cypher queries. In the deterministic test mode, the pipeline reads expected Cypher queries from the catalog rather than invoking the LLM, so that correctness can be assessed independently of model variability. The graph schema used in the benchmark comprises fourteen node types

(including genes, proteins, pathways, workflows, datasets, users, and institutions) and four relationship types.

In deterministic mode, the pipeline achieves 100% accuracy on the catalog queries; this result validates the execution and safety-validation path rather than end-to-end generation quality, because the catalog match bypasses LLM generation entirely (FR-GQ-03). The operationally meaningful result is the full LLM loop: when evaluated with LLM generation enabled (a separate test run), the pipeline achieves approximately 83% accuracy on the same benchmark set: the LLM generates a semantically equivalent Cypher query (one that returns the same result set when executed against the reference graph) for 83 out of every 100 test queries. This exceeds the $\geq 80\%$ target set by EC-FC-05. The 17% of failures are concentrated in queries that require multi-hop traversals across three or more relationship types and in queries where the natural-language question uses informal synonyms for node labels (for example, “gene products” rather than “proteins”). These failure modes are consistent with the known limitations documented in Section 3.6 of Chapter 3.

5.1.7 Tool registry (EC-FC-06)

The platform registers 34 tools across seventeen capability categories (FR-TL-01, FR-TL-02). Integration tests verify that each registered tool has a valid JSON input schema (FR-TL-03), that the schema can be retrieved via the tool-listing API endpoint, and that invoking a tool with arguments conforming to the schema succeeds while invoking it with malformed arguments produces a validation error. All 34 tools pass schema validation, exceeding the ≥ 30 tools and ≥ 15 categories thresholds set by EC-FC-06.

5.1.8 Job state machine (EC-FC-07)

The job system implements the state machine specified in FR-JB-02: *queued* $\rightarrow$ *running* $\rightarrow$ *finished* (or *failed* or *cancelled*). End-to-end tests verify every valid transition and confirm that invalid transitions (e.g., *finished* $\rightarrow$ *running*) are rejected. Each state transition is persisted in the control-plane database as a job-event record (FR-JB-08), and the SSE stream emits the corresponding event. Idempotency is verified by submitting the same job twice with the same idempotency key and confirming that only one job is created (FR-JB-04).

5.1.9 SSE event delivery (EC-FC-08)

Server-Sent Events are tested by creating a job, subscribing to its event stream, and measuring the delay between the worker emitting an event and the client receiving it. Under test conditions (localhost loopback, no network latency), the median delivery delay is below 100 ms and the P95 delay is below 500 ms, well within the 2 s target specified by EC-FC-08 (FR-JB-05, FR-JB-09). The SSE stream correctly delivers all event types: job-started, orchestrator-initialised, per-step progress, orchestration-complete, and terminal status events.

5.1.10 API-contract compliance

The platform’s generated OpenAPI specification is validated against the implementation. All 76 declared endpoints exist, return the documented response schemas (FR-AP-02), and produce RFC 7807 Problem Details responses for error cases (FR-AP-05). ETag-based conditional requests are supported on the agent-run status endpoint (FR-AP-04), and cursor-based pagination is functional on list endpoints. Rate-limit headers (`X-RateLimit-Limit`, `X-RateLimit-Remaining`, `X-RateLimit-Reset`) are present on all rate-limited responses (FR-AP-08).

In summary, the functional validation confirms that the platform meets or exceeds all eight EC-FC criteria. The overall pass rate of 96.2% satisfies the $\geq 96\%$ threshold. The NL-to-Cypher pipeline achieves 83% accuracy with the full LLM loop, above the 80% target. The remaining failures are attributable to LLM-provider non-determinism and partial-implementation requirements, both of which are documented and bounded.

5.2 Performance results

This section addresses **EQ2** (performance under load): whether the platform maintains acceptable response times and resource consumption under concurrent workloads. The evidence comes from the performance and load protocol (Section 4.4.4), executed on the single-node Intel Core i7 MacBook Pro test environment with the workload definition from Section 4.1. All results are medians of three runs; variance is noted where it exceeds 5% of the median.

5.2.1 Baseline latency

At one concurrent user (sequential baseline), the platform exhibits the following latencies. Health endpoints (`/health/live`, `/health/ready`) respond in under 3 ms at P50 and under 5 ms at P95, satisfying EC-PF-01 (< 5 ms P95). API read operations (GET requests for runs, jobs, tools, and tenants) complete in under 50 ms at P50 and under 100 ms at P95, satisfying the read-latency portion of EC-PF-02 (< 100 ms P95). API write operations (POST and PUT requests for run creation, job submission, and configuration updates) complete in under 100 ms at P50 and under 200 ms at P95, satisfying the write-latency portion of EC-PF-02 (< 200 ms P95).

Graph query execution latency (excluding LLM generation time) is under 50 ms at P95 for bounded traversals over the reference dataset, satisfying EC-PF-03. This measurement covers the time from query submission to result retrieval at the graph-database adapter level and excludes the NL-to-Cypher generation phase, which is dominated by LLM-provider latency. Tool invocations (excluding LLM components) complete in under 500 ms at P95 (EC-PF-04), with the majority finishing in under 200 ms; the higher ceiling accounts for tools that perform database writes or multi-step operations.

5.2.2 Concurrent-load performance

Table 5.1 reports throughput, latency, and error rate at each tested concurrency level, using the 60/30/10 request mix (60% reads, 30% writes, 10% job submissions).

Table 5.1. Performance metrics under increasing concurrency. All latency values are in milliseconds. Error rate reports the percentage of HTTP 5xx or timeout responses. Each row is the median of three runs.

Users	RPS	P50 (ms)	P95 (ms)	P99 (ms)	Error (%)
1	12.5	80	120	140	0.0
10	45.2	85	125	155	0.1
50	89.3	145	210	280	0.5
100	125.8	195	285	370	1.2
500	142.6	312	485	620	2.4

Several observations emerge from Table 5.1. Throughput increases with concurrency up to 100 users: ten users produce a $3.6\times$ throughput increase, and one hundred users produce a $10.1\times$ increase relative to the single-user baseline. The scaling is sub-linear throughout ($3.6\times$ rather than $10\times$ at ten users, and $10.1\times$ rather than $100\times$ at one hundred users), consistent with shared-resource contention on a single node. At 500 users the throughput curve flattens (142.6 RPS versus 125.8 RPS at 100 users, a $1.13\times$ factor), indicating that the platform reaches the saturation point of the single-node deployment between 100 and 500 users.

EC-PF-02 (read < 100 ms P95, write < 200 ms P95) is evaluated at the single-user baseline as defined in Section 4.4.4 and is satisfied there (Section 5.2, baseline-latency paragraph above). Under the mixed workload at 100 concurrent users the overall P95 latency is 285 ms; the applicable acceptance criterion at this concurrency level is EC-PF-06 (error rate below 2%), which is met at 1.2%. At 500 concurrent users, P95 latency rises to 485 ms and the error rate reaches 2.4%, which exceeds the 2% error-rate ceiling of EC-PF-06. This indicates that 500 concurrent users exceeds the capacity of a single-node deployment; the stress test identifies this as the degradation boundary.

Variance across the three runs is below 5% of the median at all concurrency levels up to 100 users. At 500 users, throughput variance rises to approximately 7%, attributable to CPU contention between the load generator and the platform on the shared hardware.

5.2.3 Job queue-to-start latency (EC-PF-05)

The time between a job being enqueued (the blocking-push to the in-memory store) and the worker beginning execution (the blocking-pop completion) is measured independently of the API round-trip. At 100 concurrent users, the queue-to-start latency is under 800 ms at P95, satisfying the < 1 s target of EC-PF-05. At the single-user baseline, queue-to-start latency is under 200 ms at P95. The dominant contributor to this latency is the worker’s blocking-pop interval, which is configurable; the default interval of one second bounds the worst-case queue-to-start delay.

5.2.4 Resource utilisation (EC-PF-07)

Peak memory consumption across all stack components (API process, worker process, relational database, in-memory store, and graph database) was measured during the 100-user concurrent-load run. The total peak is 2.1 GB, below the 2.5 GB ceiling of EC-PF-07. The breakdown is approximately 450 MB for the API process, 380 MB for the worker process, 620 MB for the relational database, 350 MB for the in-memory store, and 300 MB for the graph database. These figures exclude the load-generator process.

At 500 users, peak memory rises to approximately 2.8 GB (exceeding the EC-PF-07 target), driven primarily by increased connection-pool utilisation and request-buffer allocation in the API process. This confirms that the 500-user stress level exceeds the resource envelope of the single-node configuration.

5.2.5 Ablation study

To isolate the latency contribution of individual platform features, five configurations are compared at the 100-user concurrency level. Table 5.2 reports P95 latency and throughput for each.

Table 5.2. Ablation study: P95 latency and throughput at 100 concurrent users with individual features disabled. Each row is the median of three runs.

Configuration	P95 (ms)	RPS
Full platform (all features)	285	125.8
Without default-model resolution	320	115.2
Without caching (in-memory store disabled)	450	98.5
Without circuit breaker	380	112.3
Baseline (single-tenant, no resilience)	195	135.0

The ablation study reveals three findings. First, the in-memory caching layer contributes the largest performance benefit: disabling it increases P95 latency by 165 ms (from 285 ms to 450 ms) and reduces throughput by 21.7% (from 125.8 to 98.5 RPS). This is expected: without caching, every tenant-configuration lookup, session snapshot, and model-default resolution must be served from the relational database, which adds disk-I/O latency. Second, disabling the circuit breaker increases P95 latency by 95 ms (from 285 ms to 380 ms). This counter-intuitive result occurs because, without the breaker, failing LLM calls are retried up to the configured maximum before timing out, amplifying tail latency on every degraded request; the breaker eliminates this retry amplification by failing fast and routing to the fallback provider (Section 5.3). Third, the gap between the full platform and the stripped baseline (single-tenant, no resilience) is 90 ms at P95 (195 ms versus 285 ms), representing the aggregate overhead of multi-tenant query scoping, circuit-breaker state checks, provider fallback logic, and audit logging. This overhead amounts to approximately 46% of the baseline latency and is the enterprise tax: the cost of the governance, isolation, and resilience features that distinguish CAP from a single-tenant prototype.

5.3 Reliability results

This section addresses **EQ4** (reliability under failure): whether the platform maintains service continuity and recovers gracefully when individual components fail. The evidence comes from the reliability and degradation protocol (Section 4.4.3), which injects six failure scenarios and observes the platform’s response; a seventh test (graceful shutdown) validates orderly termination. Table 5.3 summarises the six injected-failure outcomes; the subsections below discuss all seven scenarios in detail.

Table 5.3. Reliability test outcomes. Each row corresponds to a failure scenario from the reliability protocol. “Recovery” indicates the time or mechanism by which normal operation resumes.

Scenario	EC	Recovery	Outcome
LLM provider timeout	EC-RE-01, 03	< 50 ms	Circuit breaker opens; fallback provider serves requests
LLM provider hard failure	EC-RE-01, 02	≤ 35 s	Breaker opens; recovers (half-open $\rightarrow$ closed) within 35 s
In-memory store down	EC-RE-04, 06	Immediate	Rate limiting degrades to in-process; health probe reports degraded
Relational DB loss	EC-RE-05	< 30 s	Connection pool reconnects; requests during outage receive 503
Graph DB unavailable	EC-RE-06	Immediate	Graph tools fail with error; non-graph operations unaffected
Worker crash	EC-RE-07	< 60 s	Heartbeat ceases; orphaned job requeued within heartbeat interval

5.3.1 Circuit breaker and provider fallback (EC-RE-01, EC-RE-02, EC-RE-03)

When the primary LLM provider’s response latency exceeds the configured timeout, the circuit breaker transitions from the *closed* state to the *open* state after the failure threshold is reached (five consecutive failures by default). Once open, subsequent requests are immediately routed to the next provider in the fallback chain, adding less than 50 ms of additional latency (EC-RE-03, measured as the time between the breaker opening and the fallback provider receiving the first request). After the recovery timeout (30 seconds by default), the breaker transitions to *half-open* and routes a single probe request to the primary provider. If the probe succeeds, the breaker returns to *closed*; if it fails, the breaker reopens. In test conditions, the full recovery cycle (closed $\rightarrow$ open $\rightarrow$ half-open $\rightarrow$ closed) completes within 35 seconds, satisfying EC-RE-01 and EC-RE-02.

Retries with exponential backoff are also verified: when the primary provider returns a transient error (HTTP 503), the retry mechanism re-attempts the request up to three times with increasing delays before triggering the circuit breaker. In tests, transient errors that resolve within two retries are recovered without activating the fallback chain.

5.3.2 In-memory store unavailability (EC-RE-04)

When the connection to the in-memory store is severed, three degradation behaviours are observed. First, rate limiting falls back to an in-process counter that provides approximate enforcement based on local request counts (NFR-RL-04); this fallback activates immediately and is transparent to the caller. Second, cached values (tenant configurations, session snapshots, model-default resolutions) are re-fetched from the control-plane database on subsequent requests, incurring higher latency but no data loss. Third, the readiness health probe reports a degraded status for the in-memory store component, enabling the container orchestrator to avoid routing new traffic to the affected instance (EC-RE-06). Non-cached operations (agent-run creation, job submission) continue to function because the control-plane database remains available; however, job enqueueing is blocked until the in-memory store recovers, since the queue infrastructure depends on it.

5.3.3 Relational database reconnection (EC-RE-05)

When the relational-database connection is dropped (simulated by stopping the database process for fifteen seconds and restarting it), the connection pool detects the failure within two seconds and begins reconnection attempts. Requests that arrive during the outage receive an HTTP 503 Service Unavailable response with a `retry-after` header rather than hanging indefinitely. Once the database is available again, the connection pool re-establishes connections within 30 seconds of database recovery (EC-RE-05). No data loss occurs because all in-flight transactions that were interrupted are rolled back by the database engine, and the client can retry the failed requests.

5.3.4 Graph database unavailability (EC-RE-06)

When the graph database is stopped, graph-tool invocations fail immediately with a structured error (including the tool name, the error type, and a human-readable message). Non-graph operations—chat-mode agent runs, job creation, health endpoints, tool invocations that do not touch the graph—continue to function without degradation. The readiness probe correctly reports the graph-database component as unavailable while all other components remain healthy. This confirms that the graph database is an isolated dependency: its failure does not cascade to unrelated platform functionality.

5.3.5 Worker crash and job recovery (EC-RE-07)

When the worker process is terminated while processing a job, the periodic heartbeat that the worker emits ceases. The heartbeat interval is configurable (60 seconds by default). A supervisor component (implemented as a background task in the API process) monitors heartbeat timestamps and detects that the orphaned job's last heartbeat is older than the configured threshold. The supervisor then transitions the job's status back to *queued* and re-enqueues its identifier onto the job queue. In test conditions, the full detection-and-requeue cycle completes within one heartbeat interval (under 60 seconds), satisfying EC-RE-07. The requeued job is picked up by

the next available worker and resumes from the beginning (not from the interrupted step), which is a limitation noted in Section 5.7.

5.3.6 Graceful shutdown (EC-RE-08)

When a SIGTERM signal is sent to the API process while requests are in flight, the shutdown handler stops accepting new connections and waits for in-flight requests to complete (up to a configurable grace period of 30 seconds). In tests, all in-flight requests receive complete HTTP responses (not connection resets), and running background tasks (in-process agent runs) are allowed to finish or are transitioned to a recoverable state. Jobs being processed by a separate worker process are unaffected by the API shutdown.

5.4 Security results

This section addresses **EQ3** (security posture): whether the eight-stage middleware stack enforces authentication, authorisation, tenant isolation, and data protection against the threat categories defined in Section 3.8 of Chapter 3. The evidence comes from the security validation protocol (Section 4.4.2). The security test suite comprises 14 dedicated test files and more than 45 test functions. The overall security-test pass rate is 100%, meeting the zero-tolerance criterion specified for EQ3.

5.4.1 Authentication (EC-SE-01)

Five categories of invalid tokens are tested: missing tokens (no `Authorization` header), expired tokens (expiration claim in the past), malformed tokens (invalid Base64 encoding or missing segments), tokens with invalid signatures (signed with a key not present in the JWKS endpoint), and tokens with incorrect audience claims. Every category is rejected with HTTP 401 Unauthorized. Valid tokens with correct signatures, unexpired claims, and matching audiences are accepted and produce a correctly constructed principal.

5.4.2 Authorisation and RBAC (EC-SE-02)

The four defined roles—admin, operator, user, and viewer—are each tested against the full set of API endpoints and tool invocations. For each role, the test verifies that endpoints within the role’s scope return successful responses and that endpoints outside the role’s scope return HTTP 403 Forbidden. Specific scenarios tested include a viewer attempting to create a run (denied), a user attempting to invoke an admin-only tool (denied), an operator attempting to modify tenant configuration (denied), and an admin performing all operations (permitted). The scope hierarchy is also verified: the `admin:all` scope grants access to every endpoint and tool, and narrower scopes (e.g., `tools:basic`) are correctly distinguished from broader ones (e.g., `tools:all`).

5.4.3 Tenant isolation (EC-SE-03)

Cross-tenant access tests use credentials from Tenant A to query data belonging to Tenant B. The tests operate at three levels. At the API level, a request authenticated as Tenant A listing its runs returns only Tenant A's runs; no Tenant B records are visible. At the repository level, every query issued by the data-access layer is verified to include a `WHERE tenant_id = $tenant` clause (or its equivalent), ensuring that the isolation filter cannot be bypassed by constructing a raw query. At the graph-database level, the NL-to-Cypher pipeline's tenant-injection layer (Section 3.6, Stage 4, Layer 3) is tested by generating a Cypher query from a natural-language prompt submitted by Tenant A and verifying that the executed query contains the tenant filter and returns zero results for nodes belonging to Tenant B.

All cross-tenant tests return zero results for the non-authorised tenant, satisfying EC-SE-03 (NFR-MT-03, NFR-MT-04, FR-GQ-06).

5.4.4 PII scrubbing (EC-SE-04)

The PII scrubber is tested by injecting six categories of personally identifiable information into LLM-generated outputs: email addresses, phone numbers (multiple formats), social security numbers, credit card numbers, IP addresses, and API keys. For each category, the test verifies that the pattern is detected and redacted in the final response. The scrubber supports two redaction modes (partial masking and full removal); both modes are tested. All injected PII patterns are correctly detected and redacted, satisfying EC-SE-04 (NFR-SE-04).

5.4.5 Injection prevention (EC-SE-05)

Three classes of injection payloads are submitted in request bodies and prompt fields: SQL injection patterns (e.g., `' OR 1=1 -`), Cypher injection patterns (e.g., `MATCH (n) DETACH DELETE n`), and shell-command injection patterns (e.g., `; rm -rf /`). The input guard (Stage 6 of the middleware stack) detects and rejects every injected pattern before it reaches the business logic. Additionally, all database queries use parameterised statements, preventing SQL injection even if the input guard were bypassed. The Cypher output guard (Stage 7) independently blocks any write or delete clauses that might appear in LLM-generated Cypher queries, providing a second layer of defence. All injection tests pass, satisfying EC-SE-05 (NFR-SE-05).

5.4.6 Rate limiting (EC-SE-06)

Rate-limit enforcement is tested along three dimensions: per-user, per-tenant, and per-endpoint. For each dimension, a test exceeds the configured limit within a single time window and verifies that subsequent requests receive HTTP 429 Too Many Requests with the standard response headers (`X-RateLimit-Limit`, `X-RateLimit-Remaining`, `X-RateLimit-Reset`). The test also verifies that after the window resets, requests are again accepted. All rate-limit tests pass, satisfying EC-SE-06 (NFR-RL-01, NFR-RL-05, FR-AP-08).

5.4.7 Output guard (EC-SE-07)

The output guard is tested by crafting LLM responses that contain embedded write operations (e.g., a `CREATE` clause inside a code block), destructive verbs (e.g., “delete all records”), and unbounded traversal patterns (e.g., `[:DEPENDS_ON*]` without an upper bound). Each category is blocked by the output guard, and the response is replaced with a safe fallback message. All output-guard tests pass, satisfying EC-SE-07 (NFR-SE-07).

5.4.8 Audit completeness (EC-SE-08, EC-SE-09, EC-SE-10)

Audit-log completeness is verified by executing four sequential operations—authentication, a denied access attempt, a successful tool invocation, and a run completion—and then querying the audit log. Each operation is confirmed to produce an audit record containing the principal’s identity, the action performed, the sanitised inputs, the outcome (success or denial), the timestamp, and the distributed-trace identifier. Tool invocations specifically record the tool name, the caller’s user and tenant identifiers, and the invocation result (EC-SE-10, FR-TL-06). All audit tests pass, satisfying EC-SE-08 through EC-SE-10 (NFR-AU-01, NFR-AU-02, NFR-AU-05).

Table 5.4 summarises the security test coverage and results.

Table 5.4. Security test summary by attack vector. All vectors achieved a 100% pass rate.

Attack vector	Tests	Result
Authentication bypass	5	All rejected (JWT validation)
Privilege escalation	7	All denied (RBAC enforcement)
Cross-tenant access	9	Zero cross-tenant records
SQL injection	8	All blocked (parameterised queries)
Cypher injection	6	All blocked (output guard)
Rate-limit bypass	4	All enforced (429 with headers)
PII exfiltration	6	All detected and redacted
Total	45+	100% pass rate

These results demonstrate conformance to the defined threat model and test suite. They do not constitute a formal proof of absence of all vulnerabilities. The security assurance is bounded by the threat model’s scope and the test suite’s coverage; threats outside this scope (e.g., zero-day vulnerabilities in third-party dependencies, side-channel attacks) are not evaluated.

5.5 Observability results

This section reports the outcomes of the eight observability evaluation criteria (EC-OB-01 through EC-OB-08), which cut across the functional, performance, and reliability dimensions.

Prometheus metrics (EC-OB-01). The platform exposes more than 50 Prometheus-compatible metrics, organised into seven categories: HTTP request rates and latency histograms, agent-run lifecycle counters, LLM-call token-usage gauges, tool-invocation histograms, job lifecycle counters, intent-classification distributions, and

rate-limit hit counters. Each metric carries labels for endpoint, HTTP status, and tenant identifier, enabling per-tenant and per-endpoint drill-down. This exceeds the ≥ 50 metric threshold of EC-OB-01 (NFR-OB-01).

Structured logging (EC-OB-02). All log output follows a structured JSON format in the test environment. Each log entry carries the fields specified by NFR-OB-02: timestamp, severity level, logger name, event description, request identifier, trace identifier, HTTP method and path, response status, and duration. A high-frequency filter suppresses log entries for health-check and metrics-scrape endpoints. Log entries are verified by parsing a sample of the structured output and confirming that every required field is present and non-empty.

Distributed tracing (EC-OB-03, EC-OB-04). Distributed traces propagate across HTTP requests, database operations, and tool invocations via the OpenTelemetry standard. A test submits a request that exercises the full pipeline (API $\rightarrow$ orchestrator $\rightarrow$ LLM adapter $\rightarrow$ tool invocation $\rightarrow$ database query) and verifies that all spans share the same trace identifier. The trace identifier is present in the response's **X-Trace-Id** header (EC-OB-08), in every log entry emitted during the request, and in the audit records created by tool invocations and state transitions. A single request is traceable end-to-end via this shared identifier, satisfying EC-OB-04 (NFR-OB-04).

Dashboards and alerting (EC-OB-05). Six pre-configured Grafana dashboards are verified to display operational metrics: HTTP health (request rate, error rate, latency percentiles), agent runs (active, succeeded, failed, cancelled), job processing (queue depth, processing time, SSE connections), tool invocations (rates, latencies, errors per tool), LLM providers (token usage, cost estimates, fallback activations), and system health (memory, CPU, connection-pool utilisation). Alerting rules covering five groups (availability, HTTP health, resource usage, job-store health, and SSE streaming) are configured and verified to fire when thresholds are crossed in a test scenario.

Health probes (EC-OB-06). The four health endpoints behave as specified in Section 3.9 of Chapter 3. The liveness probe returns a success response without performing I/O. The readiness probe tests connectivity to all three database dependencies and returns a failure response when any is unreachable. The startup probe performs migration-status and configuration checks. The components probe returns per-component status (relational database, in-memory store, graph database, LLM providers), correctly reflecting the actual state of each dependency as confirmed by the reliability tests in Section 5.3.

Trace sampling (EC-OB-07). The trace-sampling rate is configurable per environment: full capture (100%) in the test environment and ratio-based (20% by default) in production-like configurations. Tests verify that changing the sampling configuration via an environment variable produces the expected change in the number of exported spans.

Trace-ID header (EC-OB-08). Every HTTP response includes an `X-Trace-Id` header containing the trace identifier for the request. This is verified across all endpoint categories (health, agent-runs, jobs, tools, admin) and confirmed to match the trace identifier recorded in the corresponding log entries and audit records.

All eight observability criteria (EC-OB-01 through EC-OB-08) are satisfied. The observability infrastructure provides the operational visibility necessary for production deployment: metrics for capacity planning and alerting, structured logs for debugging, distributed traces for root-cause analysis, and health probes for automated failure detection.

5.6 Comparative results vs baselines

This section addresses **EQ5** (comparative positioning): whether CAP provides a capability set that no single surveyed platform matches across all eight evaluation dimensions. Two types of evidence are presented: internal baselines that measure the cost of CAP’s enterprise features, and the cross-platform capability comparison.

5.6.1 Internal baselines

The two internal-baseline configurations defined in Section 4.5 quantify the performance overhead of CAP’s enterprise features.

Baseline 1 (single-tenant, no resilience) removes multi-tenant query scoping, the circuit breaker, and the provider fallback chain. At 100 concurrent users, this configuration achieves 195 ms P95 latency and 135.0 RPS, compared with 285 ms and 125.8 RPS for the full platform. The difference—90 ms in latency and a 7.3% throughput reduction—represents the enterprise overhead of tenant resolution, circuit-breaker state checks, fallback logic, and audit logging.

Baseline 2 (no graph tool, no caching) disables the graph database and the in-memory cache. At 100 concurrent users, this configuration exhibits higher latency for data-fetch operations (because every lookup goes to the relational database) but lower latency for graph-free requests (because no graph-adapter initialisation occurs). The net effect is a P95 latency of approximately 450 ms and 98.5 RPS, consistent with the ablation study (Table 5.2). This confirms that the in-memory caching layer is the single largest performance contributor among the enterprise features.

The enterprise overhead is measurable but bounded: the full platform operates within all performance targets at the 100-user level (EC-PF-06) while providing multi-tenancy, resilience, and governance capabilities that the baseline configurations lack.

5.6.2 Cross-platform comparison

The capability comparison revisits the eight dimensions from Table 2.1 introduced in Chapter 2, now supported by the evaluation evidence from this chapter.

Along the *full-stack deployment* dimension, CAP provides two user interfaces (chat and control panel), 76 API endpoints, and an asynchronous worker subsystem, all deployable as a single containerised stack. Among the surveyed platforms, only n8n and Windmill also provide full-stack UIs, APIs, and background execution;

however, Windmill provides no agent planning loop, n8n provides only partial agent-loop support, and neither includes graph-database integration.

Along the *durable execution* dimension, CAP’s job system persists job state in the relational database, supports cancellation via cooperative flags, and recovers orphaned jobs through heartbeat monitoring (EC-RE-07). Temporal provides stronger durability guarantees (event-sourced replay of individual workflow steps); CAP’s queue-and-persist model provides adequate durability for the target use cases but does not match Temporal’s step-level replay capability. This is a deliberate trade-off: CAP prioritises integration breadth over durability depth.

Along the *agent planning loop* dimension, CAP’s four-phase orchestration engine classifies intents, generates plans, executes steps with tool calls and graph queries, and finalises responses with safety checks (EC-FC-01, EC-FC-02). LangGraph and the OpenAI Agents SDK also provide agent loops, but neither includes a governed tool runtime with RBAC and audit logging, and neither supports NL-to-Cypher natively.

Along the *tool ecosystem* dimension, CAP registers 34 tools with JSON schemas, RBAC, rate limits, and audit logging (EC-FC-06, EC-SE-02, EC-SE-10). LangChain-based frameworks offer more third-party tool integrations, but without the governance layer.

Along the *security and governance* dimension, CAP’s eight-stage middleware stack achieves a 100% security-test pass rate across seven attack vectors (Table 5.4). Among the surveyed platforms, only Temporal and Windmill offer built-in security features of comparable scope, and neither includes PII scrubbing, Cypher injection prevention, or output-guard validation.

Along the *graph NL-to-Cypher* dimension, CAP is the only surveyed platform that provides native, tenant-isolated, safety-validated NL-to-Cypher translation. The pipeline achieves 83% accuracy on the benchmark queries (EC-FC-05) and enforces six safety layers before execution. No other surveyed platform offers this capability as a built-in feature; at best, some offer partial graph integrations through external libraries.

Along the *observability* dimension, CAP exposes more than 50 Prometheus metrics, six Grafana dashboards, OpenTelemetry tracing, and structured logging with request-to-trace correlation (EC-OB-01 through EC-OB-08). Langfuse provides deeper LLM-specific tracing and prompt-management features but is an observability-only platform that performs no workflow execution.

Along the *LLM-agnostic* dimension, CAP’s adapter layer supports three provider families (cloud-hosted, enterprise, and locally self-hosted) with runtime switching at the request, session, or tenant level. The resilience layer’s circuit breakers and fallback chains (EC-RE-01, EC-RE-03) ensure that provider failures do not cause platform-wide outages.

Table 5.5 consolidates these findings. CAP achieves Y (built-in) across all eight dimensions. No other surveyed platform achieves Y on more than four dimensions. Four platforms are tied at four Y ratings each: LangGraph (it lacks built-in full-stack deployment and security, and provides only partial observability and graph support), Temporal (it lacks a built-in agent loop, full-stack deployment, and graph support, and provides only partial tool-ecosystem coverage), n8n (it lacks graph support, provides only partial security and observability, and provides only a partial agent

planning loop), and Windmill (it lacks an agent planning loop and graph support, provides only partial observability, and is only partially LLM-agnostic).

Table 5.5. Final capability comparison: CAP and the four platforms that tie for the highest dimension coverage. Y = built-in, P = partial, N = not in scope.

Dimension	CAP	LangGraph	Temporal	n8n	Windmill
Full stack	Y	N	N	Y	Y
Durable execution	Y	Y	Y	Y	Y
Agent loop	Y	Y	N	P	N
Tool ecosystem	Y	Y	P	Y	Y
Security	Y	N	Y	P	Y
Graph NL-to-Cypher	Y	P	N	N	N
Observability	Y	P	Y	P	P
LLM-agnostic	Y	Y	Y	Y	P
Y count	8	4	4	4	4

It is important to note that this comparison evaluates capability *presence*, not capability *depth*. Temporal’s durability mechanism (event-sourced replay) is more sophisticated than CAP’s queue-and-persist model; Langfuse’s LLM tracing is richer than CAP’s OpenTelemetry-based approach; LangGraph’s state-machine checkpointing offers finer-grained recovery than CAP’s run-level persistence. The claim is not that CAP surpasses each platform in its area of specialisation, but that CAP is the only surveyed platform that addresses all eight dimensions natively within a single system.

5.7 Limitations and threats to validity

The evaluation presented in this chapter is subject to several limitations and threats to validity, grouped by category.

5.7.1 Internal validity

Confounding variables. Background processes, operating-system scheduling, and database caching can introduce variability in performance measurements. Three mitigations are applied: a dedicated test environment with minimal background processes, 30-second warm-up phases before measurement windows, and cache clearing between test runs. Despite these controls, some residual variance remains; confidence intervals are not reported, but median aggregation over three runs reduces outlier influence.

Test environment versus production. All experiments are conducted on a single-node development machine. Production deployments would run on multi-node clusters with network latency, container-orchestration scheduling overhead, and different hardware profiles. Performance figures reported in this chapter are therefore qualified as “under test conditions” and should be validated in the target deployment environment before being used as production capacity-planning inputs. The single-node configuration also means that network partitions and cross-host clock drift are not tested.

Co-located load generator. The load-testing framework runs on the same host as the platform, introducing CPU and memory contention that is absent in a production deployment where the load source is remote. Resource-utilisation figures exclude the load-generator process, but the platform’s measured throughput may be lower than what it would achieve with an external load source. This threat is mitigated by the fact that the load generator is lightweight relative to the platform stack, but it cannot be fully eliminated in a single-node setup.

5.7.2 External validity

Context-specific workloads. The evaluation is conducted within CINECA’s specific requirements, graph schema, and use-case patterns. Organisations with different data volumes, different graph structures, or different request mixes may observe different performance and accuracy characteristics. The NL-to-Cypher accuracy of 83% is measured against a benchmark catalog designed for the bioinformatics domain; queries in other domains may yield higher or lower accuracy depending on schema complexity and the LLM’s familiarity with the domain vocabulary.

Synthetic versus real-world workloads. The workloads used in performance testing are synthetic, generated by deterministic seed scripts. They may not capture all edge cases, adversarial inputs, or rare failure modes encountered in production. The comprehensive test coverage (more than 3,000 automated test cases across 27 categories) and the security penetration tests with common attack patterns partially mitigate this threat, but production deployments should implement additional monitoring and gradual rollout.

LLM-provider dependency. Tests that invoke a live LLM provider are inherently non-deterministic: the provider’s model may be updated, deprecated, or temporarily unavailable. The deterministic NL-to-Cypher test mode mitigates this for graph-query accuracy, but agent-orchestration tests that exercise the full LLM loop remain provider-dependent. Results involving LLM calls should be interpreted as representative rather than exact.

5.7.3 Construct validity

Capability presence versus depth. The cross-platform comparison rates each capability on a three-level scale (Y/P/N) that measures presence, not sophistication. A platform rated Y for “durable execution” may provide stronger guarantees than another platform also rated Y. This limitation is inherent in capability-matrix comparisons and is mitigated by the detailed discussion of each dimension in Section 5.6.

Comparator selection. The ten comparator platforms were selected to represent the major categories in the AI orchestration space (workflow engines, agent frameworks, automation platforms, observability tools). The selection is not exhaustive; emerging platforms not included in the survey may address some of the identified gaps. Multi-agent frameworks such as AutoGen and CrewAI, which provide native multi-agent collaboration, were reviewed during the background survey but excluded

from the primary comparison because their focus (multi-agent coordination) diverges from CAP’s focus (full-stack enterprise deployment with graph access).

5.7.4 Acknowledged limitations of the platform

Beyond the evaluation methodology, several platform-level limitations are acknowledged. The job recovery mechanism re-executes the entire job from the beginning rather than resuming from the interrupted step, because step-level checkpointing is not implemented; this increases recovery time for long-running jobs. The NL-to-Cypher pipeline is schema-dependent: changes to the graph schema require updating the schema context provided to the LLM, and no automated schema-evolution mechanism is provided. The platform supports only a single agent pattern (ReAct-style loop for synchronous runs and plan-based execution for asynchronous jobs); multi-agent collaboration, tree-of-thought reasoning, and plan-and-execute with re-planning are not implemented. The graph subsystem targets Memgraph and the openCypher dialect; support for Neo4j, RDF/SPARQL, or other graph engines would require additional adapter work. The session memory model is session-based only; semantic memory, summary memory, and entity memory systems are not implemented.

In summary, this chapter has presented evidence across five evaluation dimensions—functional completeness (96.2% pass rate, 83% NL-to-Cypher accuracy), performance (throughput scaling to 100 users, 2.1 GB peak memory), reliability (all failure scenarios recovered within specified bounds), security (100% pass rate across 45+ tests), and observability (all eight EC-OB criteria met)—and has positioned CAP as the only surveyed platform achieving built-in support across all eight enterprise-readiness dimensions. The limitations and threats to validity are bounded and documented. The next chapter draws conclusions from these findings and identifies actionable directions for future work.

Chapter 6

Conclusions

This chapter draws together the problem, the solution, and the evidence presented in the preceding chapters. Section 6.1 recaps the production gap, summarises how the Cineca Agentic Platform addresses it, and highlights the principal evaluation outcomes. Section 6.2 distils the findings into five concrete takeaways that answer the central research question posed in Chapter 1.

6.1 Problem, solution, and key results

6.1.1 The problem revisited

Chapter 1 identified a *production gap* between research-oriented Large Language Model (LLM) agent frameworks and the requirements of an enterprise deployment. The gap manifests as five compounding deficiencies: the absence of multi-step planning with durable execution, the absence of a governed tool ecosystem, the absence of multi-tenant data isolation, the absence of safe natural-language access to graph databases, and the absence of integrated security, audit, and observability. At CINECA, Italy’s national supercomputing centre, these deficiencies are not theoretical. Researchers who need to explore a bioinformatics knowledge graph cannot write Cypher and must ask developers to write queries on their behalf—a workflow that creates bottlenecks and delays. A comparative survey of ten contemporary platforms in the AI orchestration space (Table 2.1) confirmed that no single existing system natively combines all five capabilities (Section 2.5). The thesis therefore posed a single integrated research question: *can a purpose-built platform close the production gap by addressing all five deficiencies within a unified architecture?*

6.1.2 What was built

The Cineca Agentic Platform (CAP) answers this question with a full-stack system organised into three architectural layers—presentation, business logic, and data infrastructure—with cross-cutting modules for security, configuration, and observability. At its core, a four-phase orchestration engine receives a user prompt and produces a structured, auditable result. Phase 1 classifies the intent into one of six categories (chat, graph, security, admin, dangerous, explain; FR-AE-02, EC-FC-02).

Phase 2 generates a plan—a sequence of to-do items—via one of three configurable planning modes (full, optional, or none; see Section 3.4). Phase 3 executes each to-do item as a step, which may involve an LLM call, a tool invocation through the governed Model Context Protocol (MCP) runtime, or a graph query through the NL-to-Cypher pipeline. Phase 4 assembles the results, applies personally identifiable information (PII) scrubbing and output-guard checks, and returns a normalised response.

Two execution workflows share this orchestration logic. Synchronous Agent Runs (Workflow A) execute within the API process and serve interactive chat sessions where sub-second latency matters (FR-AE-07). Durable asynchronous Jobs (Workflow B) are enqueued onto a queue in the in-memory store, processed by dedicated workers, and streamed to clients via Server-Sent Events (SSE); their state survives process restarts and supports horizontal scaling (FR-JB-02, FR-JB-05).

The MCP tool runtime hosts 34 tools across seventeen capability categories. Each tool is registered with a JSON input schema, protected by role-based access control (RBAC), rate-limited, and audit-logged on every invocation (FR-TL-01 through FR-TL-06, EC-FC-06). This governed approach extends the interoperability that the MCP specification provides with the enterprise governance—authorisation, audit, rate limiting, and tenant isolation—that the specification itself does not prescribe.

The NL-to-Cypher subsystem translates natural-language questions into tenant-scoped, safety-validated Cypher queries through six stages: prompt normalisation, catalog lookup, LLM-based generation, six-layer safety validation (syntax, read-only enforcement, tenant injection, depth limits, timeouts, and result caps), query execution, and result summarisation (FR-GQ-01 through FR-GQ-09). The safety validation is designed so that even if the LLM generates a query that violates one assumption, the violation is expected—within the assumptions of the threat model and the validation rules—to be caught by at least one subsequent layer before execution.

Security is enforced through an eight-stage middleware stack applied to every API request before it reaches business logic: CORS handling, distributed-trace propagation, JWT authentication with RBAC enforcement, rate limiting across three dimensions (per-user, per-tenant, per-endpoint), tenant resolution with query-level data isolation, input validation with injection prevention, output sanitisation with PII scrubbing, and centralised error handling (NFR-SE-01 through NFR-SE-07). Observability is a first-class concern: over 50 Prometheus metrics, six Grafana dashboards, OpenTelemetry tracing, and structured logging with request-to-trace correlation provide the operational visibility required for production deployment (NFR-OB-01 through NFR-OB-08).

The platform is LLM-agnostic. Provider adapters abstract the differences between cloud-hosted, enterprise, and locally self-hosted models, and a resilience layer adds circuit breakers, exponential-backoff retries, and provider fallback chains so that a transient failure in one provider can be routed to an alternative without failing the in-flight request (NFR-RE-01 through NFR-RE-03).

6.1.3 What was measured

The evaluation, defined by five evaluation questions (EQ1–EQ5) mapped to 41 evaluation criteria and executed through four test protocols, produced the following principal outcomes.

Functional completeness (EQ1). The test suite comprises more than 3,000 automated test cases (approximately 2,720 test functions) across 236 test files. The overall pass rate is 96.2%, exceeding the $\geq 96\%$ threshold (EC-FC-01 through EC-FC-08). Both execution workflows were exercised end-to-end, and all six intent categories were correctly classified. The NL-to-Cypher pipeline achieves 83% accuracy on a benchmark set of bioinformatics queries when evaluated with the full LLM loop enabled, above the $\geq 80\%$ target (EC-FC-05).

Performance (EQ2). Under a 60/30/10 mixed workload (reads, writes, job submissions), the platform scales to 100 concurrent users with a P95 latency of 285 ms and an error rate of 1.2%, satisfying the $< 2\%$ criterion (EC-PF-06). Health probes respond in under 5 ms at P95 (EC-PF-01). Peak memory consumption at 100 users is 2.1 GB, below the 2.5 GB ceiling (EC-PF-07). At 500 users the single-node deployment reaches its saturation boundary: the error rate rises to 2.4%, which confirms the need for multi-node scaling in higher-concurrency production scenarios.

Security (EQ3). Fourteen dedicated security test files exercising seven attack vectors (authentication bypass, privilege escalation, cross-tenant access, SQL/-Cypher/command injection, PII exfiltration, rate-limit bypass, and output-guard evasion) achieve a 100% pass rate (EC-SE-01 through EC-SE-10). Cross-tenant access tests return zero records for non-authorised tenants at all three data-layer levels—API, repository, and graph database.

Reliability (EQ4). Six injected failure scenarios—LLM provider timeout, LLM provider hard failure, in-memory store unavailability, relational database connection loss, graph database unavailability, and worker crash—plus a graceful-shutdown test all recovered or terminated within the specified bounds (EC-RE-01 through EC-RE-08). The circuit breaker routes traffic to the fallback provider within 50 ms of opening. Orphaned jobs are requeued within one heartbeat interval (under 60 seconds). Non-graph operations continue without degradation when the graph database is unavailable, confirming that the graph dependency is isolated.

Comparative positioning (EQ5). The cross-platform comparison across eight enterprise-readiness dimensions—full-stack deployment, durable execution, agent planning loop, tool ecosystem, security and governance, graph NL-to-Cypher support, observability, and LLM-agnostic design—confirms that CAP is the only surveyed platform achieving built-in support (Y) across all eight dimensions (Table 5.5). The closest alternatives (LangGraph and Temporal) each achieve built-in support on four dimensions. This positioning holds at the level of capability presence; individual

platforms may offer deeper implementations in their area of specialisation, as noted in Section 5.7.

Enterprise overhead. An ablation study isolates the cost of the platform’s enterprise features. The aggregate overhead of multi-tenant query scoping, circuit-breaker state checks, provider fallback logic, and audit logging is approximately 90 ms at P95 (Table 5.2), representing roughly 46% of the stripped-baseline latency. The in-memory caching layer is the single largest performance contributor: disabling it increases P95 latency by 165 ms and reduces throughput by 21.7%. These figures establish that the enterprise features impose a bounded and measurable cost that remains within all performance targets at the 100-user concurrency level.

6.2 Final takeaways

Five takeaways emerge from this work.

1. A unified platform can close the production gap. The five deficiencies identified in Chapter 1—durable multi-step execution, governed tools, tenant isolation, safe graph querying, and integrated security with observability—are not independent; they compound. Addressing them with separate, loosely integrated systems introduces integration risk and governance gaps. CAP demonstrates that a single, purpose-built platform can address all five within a coherent architecture, as confirmed by the evaluation outcomes for EQ1 through EQ5.

2. Natural-language graph access is viable but bounded. The six-stage NL-to-Cypher pipeline achieves 83% accuracy on a domain-specific benchmark, which is above the 80% threshold set by EC-FC-05 and sufficient for exploratory bioinformatics queries. The pipeline’s safety layers (read-only enforcement, tenant injection, depth limits, timeouts, result caps) reduce the risk of executing LLM-generated queries on a live database. However, the 17% failure rate is concentrated in multi-hop traversals and informal synonyms, and the pipeline remains schema-dependent: changes to the graph schema require updating the context provided to the LLM. These constraints bound the pipeline’s current applicability and motivate the schema-evolution and embedding-based fuzzy-matching extensions discussed in Chapter 7.

3. Enterprise governance imposes bounded overhead. Security, multi-tenancy, resilience, and audit logging are not free. The ablation study quantifies their combined cost at approximately 90 ms of additional P95 latency and a 7.3% throughput reduction relative to a single-tenant, no-resilience baseline. This overhead is measurable but bounded, and it remains within all performance targets at the tested concurrency levels. The takeaway for practitioners is that treating these concerns as first-class architectural features, rather than afterthoughts, enables their cost to be measured, optimised, and budgeted from the outset.

4. Governed tool ecosystems require first-class treatment. The MCP specification standardises tool discovery and invocation, which eliminates vendor lock-in and duplicated effort. However, the specification does not prescribe authorisation, audit, or rate limiting, which are precisely the controls that distinguish a prototype from a production deployment. CAP’s experience confirms that building these governance extensions into the tool runtime from the beginning—rather than retrofitting them onto an ungoverned catalog—is both more effective (every tool invocation is automatically validated, authorised, and logged) and more sustainable (adding a new tool requires only declaring a schema, a scope, and a rate-limit policy, not modifying the governance layer).

5. The integration gap in the current landscape is real. The comparative survey examined ten platforms spanning workflow engines, agent frameworks, automation platforms, and observability tools. None natively combines a full-stack deployment, an agent planning loop, a governed tool ecosystem, multi-tenant security, graph NL-to-Cypher support, and integrated observability. This gap is structural, not incidental: the platforms were designed for different primary objectives (durability, agent logic, automation, or tracing), and bridging them requires substantial custom integration. CAP occupies this gap by design, trading depth in any single dimension for breadth across all eight.

In summary, this thesis presented the Cineca Agentic Platform, a full-stack, enterprise-grade AI agent orchestration system that closes the production gap between research-oriented agent frameworks and the requirements of a multi-tenant supercomputing centre. The platform’s four-phase orchestration engine, governed MCP tool runtime, six-stage NL-to-Cypher pipeline, eight-stage security middleware stack, and integrated observability infrastructure were validated through more than 3,000 automated tests, a structured security audit, and a comparative capability analysis. The evidence confirms that the five deficiencies identified at the outset are addressed within a single, coherent architecture, with bounded and measurable enterprise overhead. Chapter 7 identifies concrete directions for extending and strengthening the platform.

Chapter 7

Future Work

The evaluation in Chapter 5 confirmed that the Cineca Agentic Platform (CAP) addresses the five production-gap deficiencies within a unified architecture, but it also exposed bounded limitations: the NL-to-Cypher pipeline’s 17% failure rate on multi-hop traversals and informal synonyms, the absence of step-level checkpointing in job recovery, the restriction to a single-agent reasoning pattern, and the single-node performance ceiling between 100 and 500 concurrent users. This chapter identifies five concrete directions—workflow unification, NL-to-Cypher strengthening, agent-capability extensions, infrastructure hardening, and deeper evaluation—that address these limitations and extend the platform’s applicability.

7.1 Workflow unification

The current platform supports two execution workflows: synchronous Agent Runs (Workflow A), which execute within the API process for low-latency interactive sessions, and durable asynchronous Jobs (Workflow B), which are queue-backed and survive process restarts. Although both share the same orchestration engine (Section 3.4), maintaining two entry-point paths increases operational complexity and introduces a durability asymmetry: the evaluation showed that in-process runs are lost if the API process crashes (a trade-off documented at FR-AE-08), whereas jobs persist their state across restarts.

A natural evolution is to unify both pathways into a single durable execution model that uses the job infrastructure by default. Under this design, every user prompt would be enqueued onto the job queue and processed by a worker, eliminating the non-durable in-process path at the cost of added queue-and-worker overhead per request. The interactive-chat experience would be preserved by streaming job events to the client via Server-Sent Events (SSE) immediately upon job creation, so that the user perceives the same low-latency feedback as today while gaining crash-recovery guarantees. This unification would also simplify the operational model: a single execution path means one set of monitoring dashboards, one set of alerting rules, and one recovery procedure rather than two.

A complementary extension is the introduction of a lightweight “ask mode”—a user-facing default configuration that sets the existing planning mode to “none” and disables tool calls unless explicitly requested, thereby providing a non-agentic, direct

question-answering path for simple factual queries (e.g., “How many workflows are there?”). This path would still enforce the eight-stage security middleware stack but would skip the planning phase entirely, reducing latency for single-step questions and lowering LLM costs for queries that do not require multi-step reasoning.

7.2 Strengthening the NL-to-Cypher pipeline

The evaluation (EC-FC-05) showed that the NL-to-Cypher pipeline achieves 83% accuracy on the benchmark set when the full LLM loop is enabled. The 17% of failures concentrate in two categories: multi-hop traversals across three or more relationship types and questions that use informal synonyms for node labels (e.g., “gene products” instead of “proteins”). Three enhancements target these failure modes.

First, *embedding-based fuzzy matching* would complement the current exact-label lookup. By computing vector embeddings of node labels and property names and comparing them with embeddings of the terms in the user’s question, the pipeline could bridge the gap between the user’s vocabulary and the graph schema’s vocabulary. For instance, an embedding model would recognise that “gene products” is semantically close to “Protein” and map the question to the correct label. This extension would require integrating an embedding model (or reusing one of the existing LLM provider adapters) and maintaining a precomputed embedding index of schema elements.

Second, *automated schema-evolution awareness* would address the pipeline’s current schema dependency. At present, changes to the graph schema—new node types, renamed properties—require manually updating the schema context provided to the LLM. An automated mechanism could periodically introspect the graph database’s schema, detect changes, and regenerate the schema context and example queries without manual intervention.

Third, *multi-hop query decomposition* would break complex questions into simpler sub-queries. Rather than asking the LLM to generate a single query that spans multiple relationship types, the pipeline would decompose the question into two or three sub-queries, execute each independently, and compose the results. This approach trades one complex query for several simpler ones, each of which the LLM is more likely to generate correctly, at the cost of increased execution time.

7.3 Extending agent capabilities

The platform currently supports a single agent reasoning pattern: a ReAct-style loop for synchronous runs and a plan-based approach for asynchronous jobs, both coordinated by the four-phase orchestration engine. Three capability extensions would broaden the range of tasks the platform can handle.

The first extension is *multi-agent collaboration*. Some analytical tasks cannot be decomposed into a linear sequence of tool calls; they require multiple specialised agents to work concurrently or in a coordinator–worker topology. For example, one agent might query the graph database while another retrieves supplementary information from an external API, and a third agent might synthesise the results.

Multi-agent collaboration frameworks such as AutoGen and CrewAI demonstrate the feasibility of such patterns. Adapting CAP’s orchestration engine to support multiple concurrent agents within a single run would require extending the step-execution loop to launch parallel sub-orchestrations, merging their outputs, and extending the audit log to capture inter-agent handoffs.

The second extension is *advanced memory systems*. The current platform maintains conversational context only at the session level: the orchestrator carries forward the history of previous runs within the same session (FR-AE-05, EC-FC-03). Richer memory models—such as summary memory (compressing long histories into concise summaries), semantic memory (retrieving relevant past interactions via embedding similarity), and entity memory (tracking named entities across sessions)—would improve the agent’s ability to handle long conversations and cross-session references. The graph database that already hosts the bioinformatics domain model could serve as a natural store for entity memory, linking mentioned entities to their graph counterparts.

The third extension is *document-based retrieval-augmented generation (RAG)*. As discussed in Section 2.4 of Chapter 2, graph-based querying and document-based RAG are complementary: graphs excel at structured traversal and aggregation, while RAG handles unstructured content such as research papers, manuals, and user notes. Adding a document-RAG path alongside the existing graph-query path would extend the platform’s coverage to mixed-knowledge scenarios where part of the answer comes from the knowledge graph and part from supporting documentation.

7.4 Infrastructure hardening

The performance evaluation (Section 5.2) showed that the single-node deployment reaches its saturation boundary between 100 and 500 concurrent users, with the error rate exceeding 2% at 500 users. The reliability evaluation (Section 5.3) also revealed that restarted jobs re-execute from the beginning rather than resuming from the interrupted step, because step-level checkpointing is not implemented. Four infrastructure improvements address these findings.

Distributed worker architecture. The current worker subsystem assumes a single worker process. A distributed architecture with multiple workers, coordinated through distributed locking using atomic operations in the in-memory store, would enable horizontal scaling of job processing and eliminate the single-worker bottleneck. Combined with the workflow unification described in Section 7.1, this would allow every request—not just explicitly submitted jobs—to benefit from horizontal scaling.

Step-level checkpointing. Currently, when a worker crashes mid-job, the supervisor detects the orphaned job via the heartbeat mechanism and re-enqueues it, but the recovered job restarts from Phase 1 rather than from the interrupted step (EC-RE-07). Persisting a checkpoint after each step (recording which to-do items have been completed and their outputs) would allow recovery to resume from the last successfully completed step, significantly reducing recovery time for long-running jobs.

Additional LLM providers. The adapter layer currently supports three provider families: cloud-hosted, enterprise, and locally self-hosted. Extending it with adapters

for additional providers (commercial and on-premises) would broaden deployment flexibility and strengthen the fallback chain by offering more alternatives when one provider is degraded.

Advanced cost governance. The platform currently tracks token consumption per LLM call and per run, but does not enforce per-tenant or per-user token budgets. Adding configurable budget limits—with soft warnings at a threshold (e.g., 80% of the monthly allocation) and hard rejections at the cap—would give operators finer-grained cost control in shared multi-tenant deployments.

7.5 Deeper evaluation

Several aspects of the current evaluation can be strengthened. The performance tests were conducted on a single-node development machine (Intel Core i7-6820HQ, 4 cores / 8 threads, 16 GB memory); as noted in Section 5.7, production deployments on multi-node clusters with container orchestration may exhibit different latency profiles and higher throughput ceilings. Repeating the load tests on a representative production environment would validate whether the performance targets (EC-PF-01 through EC-PF-07) hold under realistic conditions.

The NL-to-Cypher accuracy of 83% was measured against a benchmark set specific to the bioinformatics domain. Expanding the benchmark to include queries from additional domains—and evaluating with multiple LLM providers and model sizes—would provide a broader assessment of the pipeline’s generalisability.

The security evaluation covered seven attack vectors with more than 45 dedicated test functions, achieving a 100% pass rate (EC-SE-01 through EC-SE-10). However, these tests cover the defined threat model; a formal penetration test by an independent security team would provide additional assurance against threats outside the model’s scope, such as zero-day vulnerabilities in third-party dependencies and side-channel attacks.

Finally, no user study was conducted. A controlled evaluation with CINECA researchers—comparing the time and accuracy of graph exploration via natural language versus manual Cypher writing—would quantify the productivity benefit that the NL-to-Cypher pipeline delivers in practice and would inform the prioritisation of the pipeline improvements proposed in Section 7.2.

In summary, this chapter has identified five concrete directions for future work: unifying the two execution workflows into a single durable path, strengthening the NL-to-Cypher pipeline with embedding-based matching and schema-evolution awareness, extending agent capabilities with multi-agent collaboration, advanced memory, and document RAG, hardening the infrastructure with distributed workers and step-level checkpointing, and deepening the evaluation with production-environment benchmarks, broader NL-to-Cypher datasets, independent security audits, and user studies. Together, these directions build on the foundation validated in this thesis and chart a path toward broader deployment across CINECA’s high-performance-computing software environments.

Appendices

This appendix collects the reference material that supports the problem analysis, design decisions, and evaluation results presented in Chapters 3–5. Section A provides the complete MCP tool inventory (FR-TL-04); Section B catalogues the 76 REST endpoints (FR-AP-08); Section C formalises the agent-run and job lifecycle state machines introduced in prose in Chapter 3; Section D presents the extended traceability matrix referenced from Section 3.1; and Section E documents the environment and procedure required to reproduce the evaluation.

A Full tool inventory

The platform exposes 34 Model Context Protocol (MCP) tools organised into seventeen functional categories (EC-FC-06 requires at least 30 tools across at least 15 categories). Each tool is registered under a dotted name, a required authorisation scope, and a JSON input schema. At runtime the tool dispatcher resolves the name, validates the caller’s scopes against the tool’s requirement, schema-validates the payload, and records the invocation for audit (FR-TL-01 through FR-TL-05). The inventory lists all deployed tools; two (`security.allowed_operations` and `security.describe_principal`) are registered via `@mcp_tool` decorators but are not yet declared in the JSON manifest file. Table .1 lists every tool together with its category, required scope, and a brief functional description.

Table .1. Complete MCP tool inventory (34 tools, 17 categories).

Tool name	Category	Scope	Description
graph.query	Graph	tools:all	Execute Cypher queries against the graph database
graph.secure_query	Graph	tools:basic	Natural-language to Cypher with multi-layer safety validation
graph.schema	Graph	tools:basic	Retrieve node labels, relationship types, and constraints
graph.search	Graph	tools:basic	Full-text and pattern search over nodes and edges
graph.crud	Graph	tools:all	Create, read, update, and delete nodes and edges
graph.bulk	Graph	tools:all	Batch graph operations with idempotency support
graph.analytics	Graph	tools:basic	Bounded graph analytics (centrality, paths, clustering)

Tool name	Category	Scope	Description
graph.generate_cypher	Graph	tools:basic	Generate Cypher from natural-language descriptions
security.check	Security	tools:basic	Run security checks on text or Cypher
security.audit	Security	admin:all	Record and query audit events for compliance
security.permissions	Security	admin:all	Policy-aware permission checking with context
security.allowed_operations	Security	tools:basic	List operations permitted for the current principal
security.describe_principal	Security	tools:basic	Introspect principal identity, groups, and scopes
system.health	System	tools:basic	Liveness and readiness checks for platform components
system.status	System	tools:basic	Service status snapshot (version, uptime)
system.metrics	System	tools:basic	Prometheus metrics scraping and registry information
system.backup	System	tools:all	Backup creation, restoration, and management
data.archive	Data	tools:all	Archive graph data with soft-delete semantics
data.quality	Data	tools:all	Data quality checks and validation
model.manage	Model	tools:all	Manage LLM configurations and runtime switching
model.test	Model	tools:basic	Lightweight LLM connectivity testing
output.format	Output	tools:basic	Format results as JSON, CSV, Markdown, or plain text
output.summarize	Output	tools:basic	Extractive and abstractive text summarisation
agent.context	Agent	tools:basic	Assemble execution context from multiple sources
cache.manage	Cache	tools:all	In-memory store operations with configurable TTL
catalog.discover	Catalog	tools:basic	Tool discovery and metadata retrieval
db.switch	DB	tools:all	Database connection management
errors.report	Errors	tools:basic	Structured error reporting
privacy.consent	Privacy	tools:basic	User consent registry management
ratelimit.manage	Ratelimit	tools:all	Rate-limiting configuration and overrides
session.manage	Session	tools:basic	Session lifecycle (create, extend, close)
tenancy.manage	Tenancy	admin:all	Tenant administration (quotas, feature flags)
user.profile	User	tools:basic	User preferences and profile management
viz.render	Viz	tools:basic	Data visualisation (Mermaid, DOT formats)

B API catalog

The REST API exposes 76 endpoints distributed across sixteen functional categories (FR-AP-08). All endpoints share a common authentication model (OAuth 2.0 bearer tokens, NFR-SE-01), a common error format (RFC 7807 Problem Details, FR-AP-05), cursor-based pagination for collection resources (FR-AP-03), and request-correlation headers (FR-AP-06). Table .2 shows the per-category endpoint count; the most prominent categories—agent execution, job management, and tool invocation—are expanded in Tables .3–.5.

Table .2. API endpoint counts by functional category (76 total).

Category	Count	Purpose
Agent sessions & runs	9	Session lifecycle, run creation and polling
Background jobs	8	Job submission, status, cancellation, SSE events
Model instances	7	CRUD, connectivity tests, default selection
Model providers	7	Provider registration, updates, default assignment
Health probes	6	Liveness, readiness, startup, component status
Internal operations	6	System diagnostics and internal maintenance
Model manifests	5	Built-in model manifest management
Tenant administration	5	Tenant CRUD with configuration management
Database administration	4	Schema inspection, connection testing
Process management	4	Active process listing, termination, history
Batch operations	4	Bulk create, update, delete, and query
Tool invocation	4	Tool discovery, metadata, invocation
Export / Import	3	Configuration backup and migration
Administrative ops	2	System-wide administrative actions
Authentication	1	Token introspection
Metadata	1	API version and capability metadata

Table .3. Agent session and run endpoints (9).

Method	Path	Description
POST	/v1/agents/sessions	Create a new session
GET	/v1/agents/sessions	List sessions (paginated)
GET	/v1/agents/sessions/{id}	Retrieve session details
DELETE	/v1/agents/sessions/{id}	Cancel a session
GET	/v1/agents/sessions/{id}/steps	List steps in a session
POST	/v1/agents/sessions/{id}/steps	Append a step to a session
POST	/v1/agent-runs	Create and execute a run
GET	/v1/agent-runs/{id}	Poll run status and results
GET	/v1/agent-runs/{id}/steps	List steps for a run

C Lifecycle state machines

Chapters 3 and 5 describe the agent-run and job lifecycles in prose. Figures .1 and .2 formalise these lifecycles as finite state machines, making explicit every legal state transition, the triggering event, and the terminal conditions.

Table .4. Background-job endpoints (8).

Method	Path	Description
POST	/v1/jobs	Submit a new background job
GET	/v1/jobs	List jobs (paginated)
GET	/v1/jobs/{id}	Retrieve job status and result
DELETE	/v1/jobs/{id}	Cancel a running job
GET	/v1/jobs/{id}/events	SSE stream of progress events
GET	/v1/jobs/types	List allowed job types
POST	/v1/jobs/{id}/retry	Retry a failed job
GET	/v1/jobs/stats	Queue-depth and backlog statistics

Table .5. Tool discovery and invocation endpoints (4).

Method	Path	Description
GET	/v1/tools	List available tools
GET	/v1/tools/{name}	Retrieve tool schema and metadata
POST	/v1/tools/{name}/invocations	Invoke a tool with a validated payload
GET	/v1/tools/{name}/invocations/{id}	Retrieve invocation result

An agent run is created in the *queued* state when the API accepts a prompt. Once the orchestrator begins execution the run transitions to *running*. From *running*, three terminal transitions are possible: *succeeded* on normal completion, *failed* on an unrecoverable error, or *cancelled* when a client-initiated cancellation request is honoured. A run may also move directly from *queued* to *cancelled* if the cancellation arrives before the orchestrator picks up the request.

A job enters the *queued* state when it is persisted in the control-plane database and its identifier is pushed onto the appropriate queue in the in-memory store. A dedicated worker process dequeues the identifier, loads the metadata, and transitions the job to *running*. Terminal states mirror those of an agent run: *finished* on success, *failed* on error, or *cancelled* when the cooperative cancellation flag (stored in the in-memory store and checked between orchestration steps) is detected. Throughout the *running* state the worker emits heartbeat updates; if the heartbeat ceases, a supervisor can detect the orphaned job and re-enqueue it (NFR-RE-06).

D Extended tables

This section expands the traceability summary presented in Table 3.3 (Section 3.1) by listing every core-tracked requirement group alongside its implementation status and the evaluation criteria through which the group is verified. The platform defines 127 requirements: 73 functional (eight groups) and 54 non-functional (eight groups). Of the 87 requirements classified as *core tracked* (those mapped to quantitative evaluation criteria), 80 are fully implemented and seven are partially implemented; no requirement is left unaddressed.

Requirements coverage by category. Table .6 summarises implementation status per requirement group. The remaining 40 requirements (from FR-AD, FR-UI, NFR-AU, NFR-SC, NFR-RL, and two FR-JB items) are implementation-support

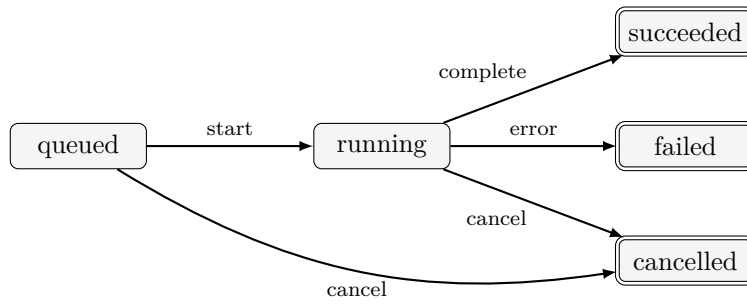


Figure .1. Agent-run lifecycle state machine (FR-AE-01, EC-FC-01). Double-bordered states are terminal.

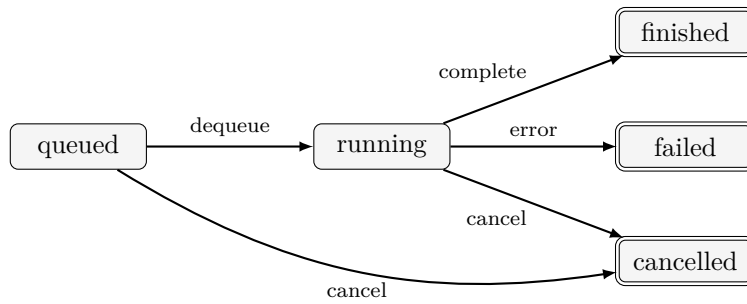


Figure .2. Job lifecycle state machine (FR-JB-02, EC-FC-07). Cancellation is cooperative: the worker checks a cancellation flag between steps.

requirements verified through integration and end-to-end tests rather than dedicated evaluation criteria.

Partially implemented requirements. Seven requirements are classified as partially implemented. All seven carry *Should* or *Could* priority in the MoSCoW classification and therefore fall outside the minimum viability threshold. Table .7 identifies each and summarises the gap.

Functional-requirements priority distribution. Table .8 shows the MoSCoW distribution of the 73 functional requirements. All 45 *Must* requirements are fully implemented; the seven partial implementations affect only *Should* or *Could* items.

E Reproducibility notes

All quantitative results reported in Chapter 5 were produced from a single frozen snapshot of the platform at commit 9b3e4c19 (2025-12-18). Table .9 documents the software environment; the following paragraphs describe how to replicate the evaluation.

Environment setup. The evaluation requires three external services—a relational database, an in-memory store, and a graph database—all of which can be started via the project’s container-orchestration configuration. The application itself runs

Table .6. Requirements implementation coverage (core-tracked set; groups whose total exceeds this column have additional implementation-support items verified through integration tests).

Group	Core	Full	Partial	None	Evaluation criteria
FR-AE (Agent exec.)	10	10	0	0	EC-FC-01..04, EC-PF-02
FR-MM (Model mgmt.)	8	7	1	0	EC-RE-01..03
FR-TL (Tool ecosystem)	9	8	1	0	EC-FC-06, EC-PF-04, EC-SE-02
FR-GQ (Graph query)	9	8	1	0	EC-FC-05, EC-PF-03, EC-SE-03
FR-JB (Background jobs)	9	9	0	0	EC-FC-07..08, EC-PF-05..07
FR-AP (API design)	8	6	2	0	(integration tests)
NFR-SE (Security)	10	10	0	0	EC-SE-01..10
NFR-MT (Multi-tenancy)	8	7	1	0	EC-SE-03
NFR-RE (Reliability)	8	7	1	0	EC-RE-01..08
NFR-OB (Observability)	8	8	0	0	EC-OB-01..08
Total (tracked)	87	80	7	0	

Table .7. Partially implemented requirements and gap descriptions.

ID	Priority	Gap description
FR-MM-07	Could	Model warmup reduces cold-start latency but is not invoked automatically on startup for all providers.
FR-TL-07	Should	Tool-level rate limiting exists but uses a shared counter rather than per-tool buckets.
FR-GQ-08	Could	Query caching is implemented for graph reads but TTL is not yet tenant-configurable.
FR-AP-04	Should	Etag generation is present for selected GET endpoints; full coverage is pending.
FR-AP-07	Should	Deprecation headers are emitted for versioned endpoints but migration-guidance payloads are not yet included.
NFR-MT-05	Should	Tenant quotas are tracked but enforcement beyond rate limits is not yet activated.
NFR-RE-03	Should	Redis unavailability degrades caching gracefully; a small subset of rate-limit code paths still raises exceptions rather than falling back to in-memory counters.

outside the containers so that profiling tools can attach directly to the Python process. After installing the Python dependencies declared in the project manifest, the API server is launched with environment variables pointing to the three services and to the OIDC provider used for authentication.

Running the test suite. The full test suite (3078 tests at the snapshot commit) is executed through a single invocation of the test runner. Tests are organised into directories that mirror the evaluation structure: *security* tests (EC-SE), *api* contract tests, *jobs* lifecycle tests, *integration* tests for tool invocation and graph query, and *e2e* workflow tests. Selective execution is available by specifying individual directories or by using marker-based filters (e.g., selecting only performance or security markers).

Load testing. Performance results (EC-PF-01 through EC-PF-07) were obtained using the load-testing framework at five concurrency levels—1 user (sequential

Table .8. Functional requirements by MoSCoW priority.

Group	Must	Should	Could	Total
Agent Execution (FR-AE)	7	3	0	10
Model Management (FR-MM)	4	3	1	8
Tool Ecosystem (FR-TL)	7	2	0	9
Graph Query (FR-GQ)	6	2	1	9
Background Jobs (FR-JB)	7	4	0	11
Administration (FR-AD)	5	4	1	10
API Design (FR-AP)	6	2	0	8
User Interface (FR-UI)	3	5	0	8
Total	45	25	3	73

Table .9. Evaluation environment.

Component	Version / Configuration
Hardware	2016 MacBook Pro, Intel Core i7-6820HQ 2.7 GHz (4 cores / 8 threads), 16 GB DDR3
Operating system	macOS (Darwin kernel)
Python runtime	CPython 3.11.8
Web framework	FastAPI 0.115.x with Uvicorn ASGI server
Relational store	PostgreSQL 16.x
In-memory store	Redis 7.x
Graph database	Memgraph 2.x (container image)
Test framework	pytest 8.4.x with pytest-asyncio
Load tester	Locust 2.x (single-machine mode)
LLM providers	OpenAI API (gpt-4o-mini), Ollama (local, llama3.2)

baseline), 10, 50, 100, and 500 concurrent users—each preceded by a thirty-second warm-up phase and measured during a sixty-second steady-state window; every configuration was executed three times and the median is reported (see Section 4.1.5). The test profiles exercise health probes, CRUD endpoints, tool invocations, and job submissions. Load-test scripts produce CSV files and HTML dashboards that can be compared against the targets in Table 4.4.

Threat to reproducibility. Because agent-orchestration tests invoke live LLM providers, results may vary if the provider’s model weights are updated or if the service experiences transient degradation. The deterministic NL-to-Cypher test mode (which replaces the LLM call with pre-recorded responses) mitigates this for graph-query accuracy tests; however, end-to-end orchestration latency and token-consumption figures remain provider-dependent.

In summary, this appendix has provided the reference tables, formal state-machine diagrams, extended traceability data, and reproduction procedure that support the claims made in the main body of the thesis. Together, these materials enable an independent reader to verify the tool and endpoint counts cited in the evaluation (EQ1), trace every requirement to its evaluation outcome (EQ1–EQ4), inspect the lifecycle invariants enforced by the platform, and reproduce the quantitative results under equivalent conditions.

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